

SHAPE UP!

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Changelog

2015-01-22	Akita Noek	Repackaged into one PDF, added table of contents, cover page, license and changelog page.
2019-12-11	Brian Dunn	Improved grid lines, embedded fonts.

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Introduction: The scope of shape

What is shape?

Strong go players have in their armoury many set patterns of play. While shape (Japanese *katachi*) could mean any pattern that regularly occurs on the go board, it is useful to restrict the idea somewhat. The most immediately valuable shape ideas are those revealing the position of *vital points*. A vital point is a key location. If you occupy it, that by itself will give you a good result, not by some magic, but because of the nature of the position.

This is very important in practical play: from the point of view of reading, playing the vital point is only looking one play deep! Strong players are able to play well without much apparent thought, simply by concentrating on correct shape (which is not to say that deep reading has no part in go).

To start out, there is a small collection of go proverbs that help one to recognise those vital points of shape. (See the proverb index on p.216.) They are heuristics, not rigorous rules, and so are best studied with their exceptions: the meta-proverb says ‘beware of applying proverbs blindly’.

How do *joseki* and *tesuji* relate to shape?

In learning the basic tactics of go, one at an early stage identifies cutting points as crucial. Some time after that, the study of *tesuji* problems shows that major tactical gains may result from certain standard plays, in particular those taking advantage of lack of liberties. Good shape plays may be less extreme or dramatic than those handled under the heading of *tesuji*, and yet still offer important advantages. Opportunities to play *tesuji* occur only a few times in a game between well-matched players, but good shape is constantly required.

Before studying shape, most players will encounter a few set openings, called in Japanese *joseki*. *Joseki* are standard sequences, including the conventional corner openings. They are patterns that have been evaluated by consensus of professionals.

A given *joseki* sequence steers a way through many possible variations, some of which are discarded as obvious tactical failures. Normally many further variations are rejected as poor shape: some of the players’ stones are

inefficient or redundant, one of the groups created has inadequate eye shape or is difficult to develop further, and so on.

Objectives of this book

To explain which points are vital in given shapes.

To show how good shape is achieved, and bad shape exploited, in fighting contexts.

To integrate shape proverbs into your knowledge of go.

To look behind the proverbs to another level of more explicit mechanism, to provide supporting material, and to explain exceptions.

To break down the barrier between *tesuji* and *joseki* points of view, connecting pure intuitions with learned knowledge.

To demystify many common *tesuji*.

To help the reader to visualise how and where a *tesuji* might happen in the future, a requirement for a *dan* player.

To discuss the choice of variation at a point in a *joseki*, when tactical reasons alone aren't a sufficient guide.

To address as we go along questions about *suji*, or correct style, covering some of the content of the many texts on '*kata* and *suji*' in the Japanese literature.

To contribute to the local, critical theory of go, by attempting a systematic listing of possibilities in a pattern, with criteria for choosing amongst them.

To develop an ingrained respect in the reader for the principles of good shape (for example: connect but remain light and flexible, don't fill in your own liberties without very good reason, develop rapidly but also take into account eye shape).

To provide a reference on shape (there are an index of shapes and a proverb index at the end, to help you refer to particular patterns).

To show in action the comparative method of go study.

What should I study at my level?

From 10 *kyu* to 5 *kyu* levels, you should probably concentrate on recognising standard shapes as they come up in your own games, or play through professional games looking for them. It may be hard to understand why mistakes in shape are bad play, until you have also studied the basic shape concepts. In particular the study of *joseki* at this point may appear to be unrewarding, plain and simple memorisation.

Problem solving, first of all about basic life-and-death and then more general tactics under the heading of *tesuji*, is likely to seem more attractive to the player interested in progressing beyond 10 *kyu*. After solving enough problems from the go literature, you should begin to find the vital points in formations. This book can serve as a reference for these standard shapes.

The extended *joseki* example studied in this Introduction is suitable for players about 5 *kyu* and stronger. You can use it as a diagnostic test for what you already know. If you don't initially get much out of it you should probably read some of the simpler sections first (see below).

The approach of this book

Books on *joseki* are arranged by variation; books on *tesuji* are organised in one of two ways: by underlying shape, or by function (as in the *Fujisawa Shuko Tesuji Dictionary*, the current standard work). None of these structures makes for readability, but they are suitable for reference works. We normally adopt a combination of shape and function approaches.

Studying this book

Some people will read this work through as a book (if you are of *dan* level you might enjoy this); perhaps alternating with a book on *tesuji*. There are five main parts, each starting on a fresh area, within which the chapters generally increase in difficulty. The parts, too, become harder as the book progresses. Each chapter is broken down into short sections dealing with a specific topic. There are also three problem sets, the third being much harder than the others.

Otherwise you may find it helpful to study one of these shape 'courses'. You can use these section lists as our indications of difficulty.

First reading (10 *kyu* course):

1.1, 1.2, 1.3. 2.1 to 2.5. 3.1, 3.4, 3.5B, G and M. 4.1, 4.2, 4.4, 4.5, 4.6. 5.1, 5.3, 5.4. Problem Set 1 first half. 7.1, 7.4, 7.5, 7.6. Problem Set 2 first six problems. 11.1. 13.1, 13.2, 13.4.

Second Reading (5 *kyu* course):

Introduction. 1.4, 1.5. 2.6. 3.2, 3.3. 4.3, 4.7, 4.8, 4.9. 5.2, 5.5, 5.6. 6.1, 6.3. Rest of Problem Set 1. 7.2, 7.3, 7.7, 7.8. 8.1, 8.2, 8.3. 9.1. Rest of Problem Set 2. Chapter 10. 11.2, 11.4. 12.4. 13.3, 13.7. 14.1, 14.2.

Third Reading:

3.5. 6.2, 6.4. 8.4, 8.5. 9.2, 9.3. 11.3, 11.5. Chapter 12. 13.5, 13.6. Chapters 14 and 15. Problem Set 3.

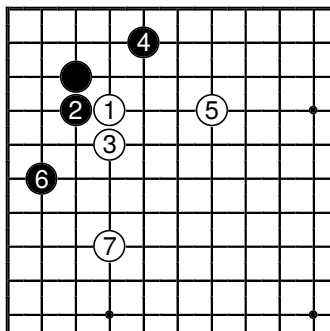
References

This book could usefully be read in parallel with *Tesuji* by James Davies (Kiseido), *Get Strong at Tesuji* by Richard Bozulich (Kiseido, this book has many examples on correct *suji*), *Tesuji and Anti-Suji of Go* by Eio Sakata (Yutopian), and *Proverbs*, Max Golem translator (Yutopian). For a general introduction to go read *Teach Yourself Go* by Charles Matthews (Hodder & Stoughton/NTC) which provides enough background to begin this book. We refer in the text to ideas of Bruce Wilcox; *EZGO – Oriental Strategy in a Nutshell* (Ki Press, ISBN 0-9652235-4-X), written with Sue Wilcox, is a representative book.

An example treated *joseki*-style

The rest of this Introduction works over a single opening pattern. This approach is typical of *joseki* books: you take a single corner opening and discuss a number of variations. You can read on to get a feel for the overall scope of shape ideas, and examples of some basic proverbs, in the context of decision-making. There's therefore the disadvantage of no single conclusion or main point. That partly explains why this isn't the way adopted in most of the rest of the book. Generally we take one pattern and look at it in various contexts. Then there is some basis for comparative reasoning, and a bit more dogmatism.

◆ Cross-references are given to the main text, in case at some point you wish to follow them up, and a number of proverbs are highlighted like *this*.



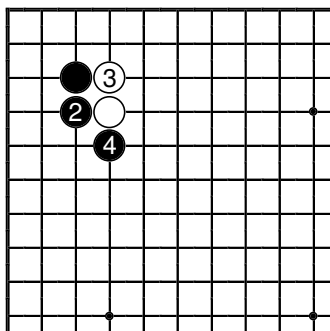
This is a regular sequence in a corner occupied first by Black, at the 3-3 point. It is relatively simple: Black's control of the corner is not contested, while White plays to avoid making a weak group. The White group then exerts influence in the centre.

It is decided early on that Black has the corner, and after that both players can be said to be 'making shape'.

Behind even such an ordinary development there may lie dozens of variations. We shall look at quite a number in the remainder of this introduction, as a way of surveying the facets of the idea of shape.

◆ Other related 3-3 patterns are to be found in 1.4, 1.5, 3.1 and 3.3.

Immediate loss of good shape

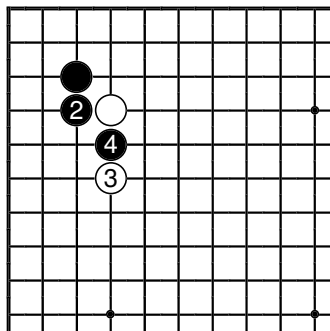


To begin with a fundamental example, White's choice of 3 here is bad shape. With 4 Black applies the proverb *play hane at the head of two stones*. After that White cannot get a good result in this part of the board.

If you commonly allow this to happen to your stones, you can probably make an instant improvement by avoiding this sort of result.

◆ This and related proverbs are studied in Chapter 4.

Playing for light shape

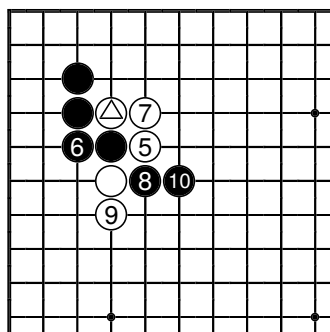
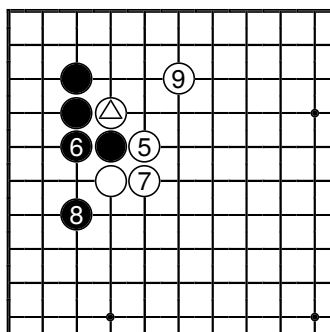


There is another recognised possibility for White 3. White can make the one-point jump played in this diagram. White must understand the purpose of this move. When Black plays 4 as shown, White will not be able to connect the two stones solidly. Therefore White 3 is a so-called *light shape*.

The main reason to prefer light shape (2.6) is that it makes defensive tasks easier. A characteristic mistake of amateur players is to make *heavy shape*: to develop groups without sufficient regard to their future defensive requirements.

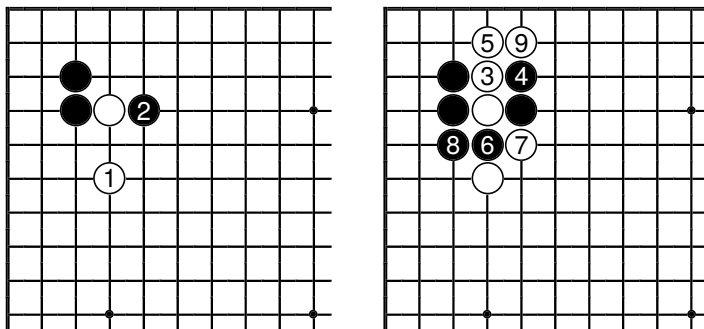
This problem manifests itself in various ways: reluctance to sacrifice stones, even those without any great strategic significance; a greedy attitude to invasions, not admitting that the opponent deserves at least some territory; reduction plays that are too deep, and which have to struggle for life; too many solid connections and groups underdeveloped because of unreasonable fear of later cuts or invasions.

Strong players are keen on sacrificing stones. They can do that successfully, for several reasons: understanding of which stones can be sacrificed, and which are essential; good technique to get the most out of a sacrifice; accurate judgement of the resulting position; and a sharp eye for later use of stones given up.

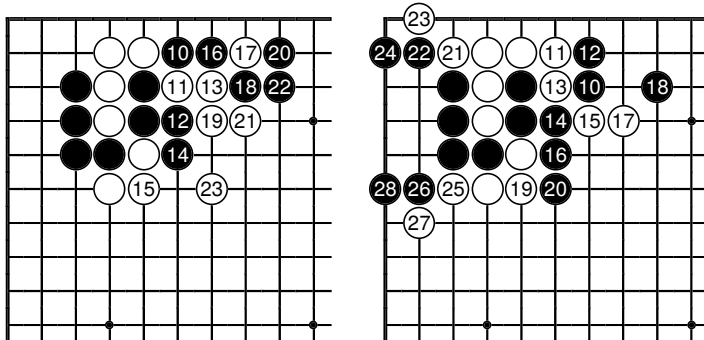


Normally White plays on 5 and 7 the outside, as in the left-hand diagram, and treats the marked white stone as a potential sacrifice. If White tried harder to save it, as in the right-hand diagram, White would immediately be involved in a difficult fight when Black cuts with 8.

Tactics for playing close



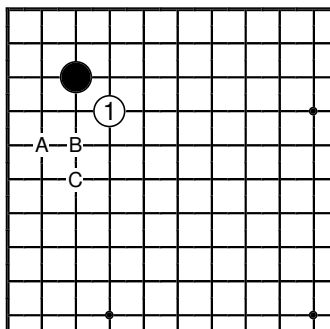
Another possibility for Black, the clamp play 2 (**left**), is not always a reliable, strong shape. (**Right**) White can try 3 and 5. What now for Black?



Black 10 in the left-hand diagram runs into immediate trouble. It is really too close. White 23 nets Black. It would be better to jump back (**right**). There the fight is more complex, but it seems White's plan is unreasonable.

◆ More about the clamp on pp.29, 31, 62.

What are my options?

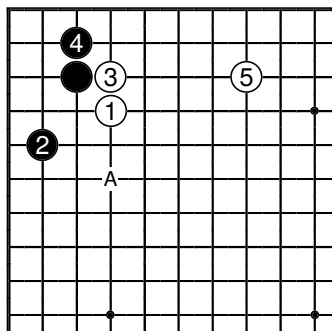


Turning back now, what choice did Black have in answering White's 'shoulder hit' play at 1? What about A, B or C instead?

These are examples of plays that are rarely considered in books. Two of them are quite bad, one of them has appeared in a top level match played by a master of the 3-3 point.

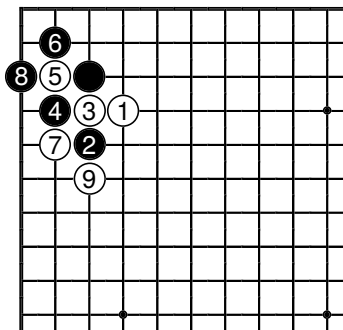
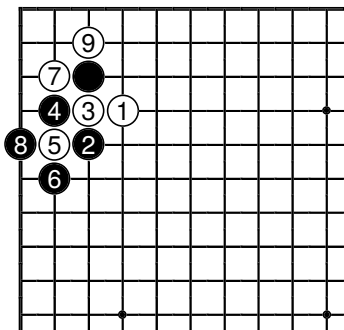
It would be an enormous task to list imaginable variants in *joseki*, or even reasonable questions to ask. One can be sure only that the plays current in professional games at a particular era have been carefully considered.

The orthodox extension



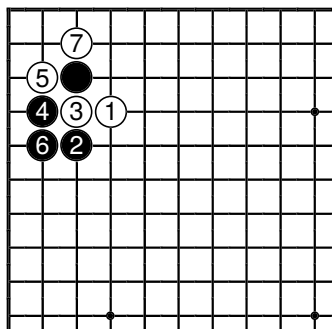
Black 2 is from professional play, though not very common. This development can be expected. White 5 *extends three from a two-stone wall*, in line with the proverb; it might also sometimes be played at A. This was Sakata 9 *dan*'s play, seen in the 1969 Judan title match against Otake. In that context White 5 wasn't possible.

Capturing a cutting stone



Choice B is bad shape. White can create two cutting points with 3. Black has no choice about playing 4, to avoid being cut cleanly in two.

White can now cut on either side, depending on the overall position. **(Left)** White can take the corner. **(Right)** With a favourable ladder White can also play for the outside. Both results are good, given that Black started here.



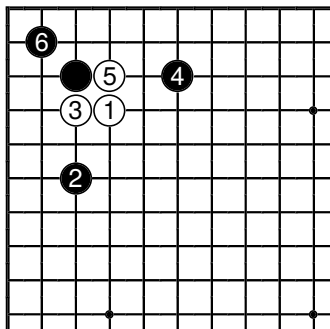
Black does well to obey the proverb *capture the cutting stone*. If Black 6 connects as shown here, the result is worse shape than capturing the stone and giving up the outside.

A single stone capture, often called *ponnuki*, may be of high value. It may build eye shape, or influence; and is normally easy to develop in more than one direction.

We have described Black's play as mistaken. So it is. Giving up the corner is a serious loss, and White's good shape is too costly.

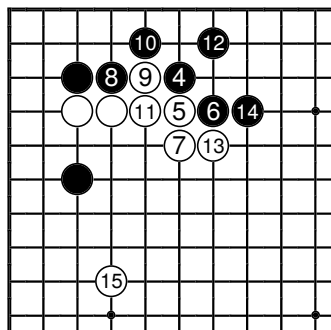
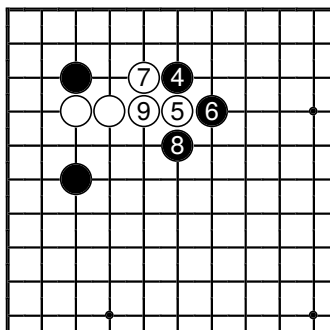
◆ The box in 1.2 discusses the proverbial value (30 points) of the *ponnuki*.

A tricky use of symmetry



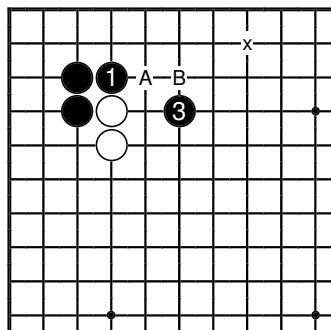
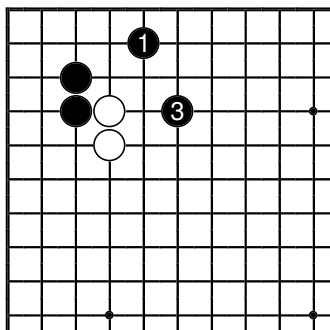
Answer C is in fact an outright trick play. It aims at this situation, in which Black 6 gains life in the corner (a case of the proverb *preserve symmetry*).

Now White cannot prevent Black's life or connection out. Black's hope is that White will get into trouble on the outside.



The *attach-block* shape (**left**) is good, in this case, if White wishes to play for territory. White can eventually take the corner, but must take great care about being shut in. Another way for White to deal with this trick play is to apply the *attach-extend* shape with 5 and 7 (**right**), and pincer with 15. However White 13 is *pushing from behind*, which is generally poor style; and Black's stone on the left side could fight back hard, at some later time.

A discussion about the future



You can't force the opponent to answer. The shape Black makes on the left with 1 and 3, when White ignores 1, is better than the shape on the right. There is an actual weak point 'x' one can see, related to White possibly playing A, which Black would like to answer solidly at B.

Part One

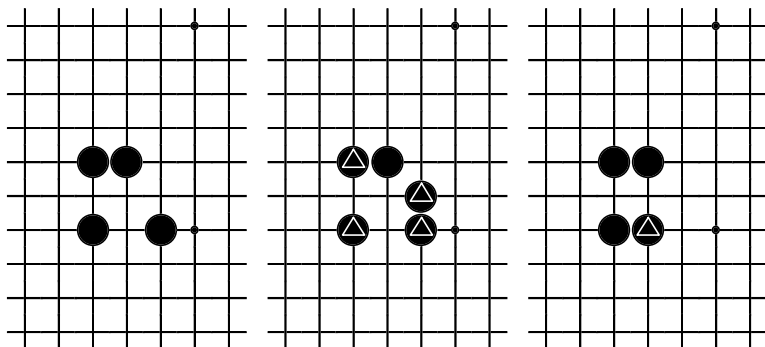
Principles of Development

Chapter One

Table Shapes

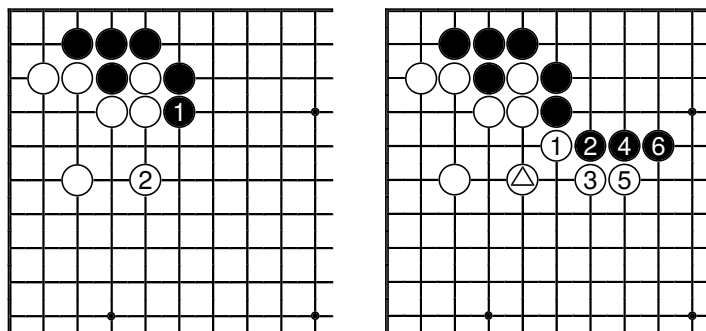
1.1 Three strong shapes

This chapter introduces a number of strong shapes. Strong shape isn't the only kind of good shape, but this is a place to start.



(Left) The basic *table* shape. **(Centre)** The *double table*, a perfect eye shape, and made up of two superposed table shapes, the second one marked.

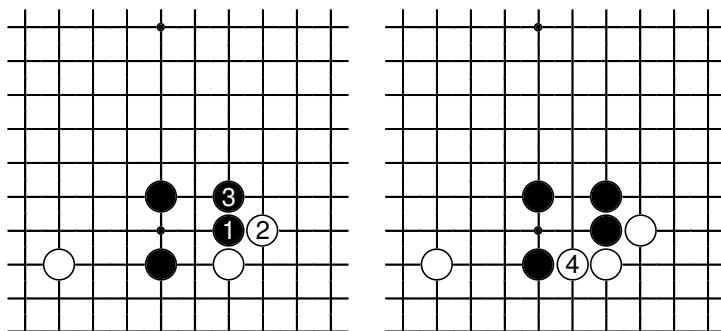
(Right) The *bamboo joint*, differing from the table shape only as marked.



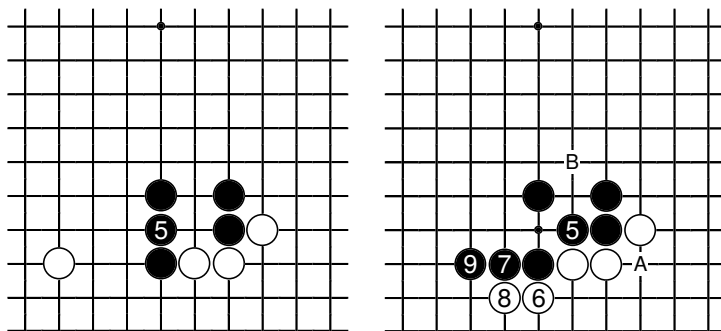
(Left) When White plays 2 here, the White group develops perfect shape.

(Right) If White's strategy requires subsequent plays pushing in the centre, they are easy to find, and the marked stone is exactly placed.

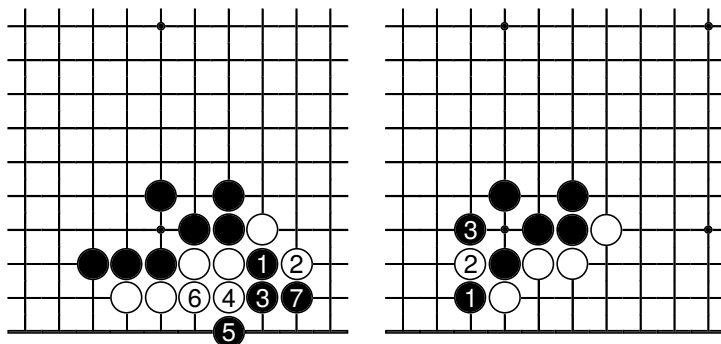
1.2 Building tables



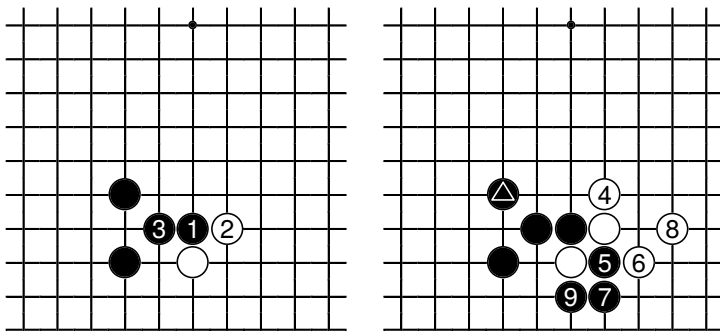
Black builds a table shape. **(Left)** Black's one-point jump emerges reinforced. **(Right)** Now White is likely to occupy the key point for eyes.



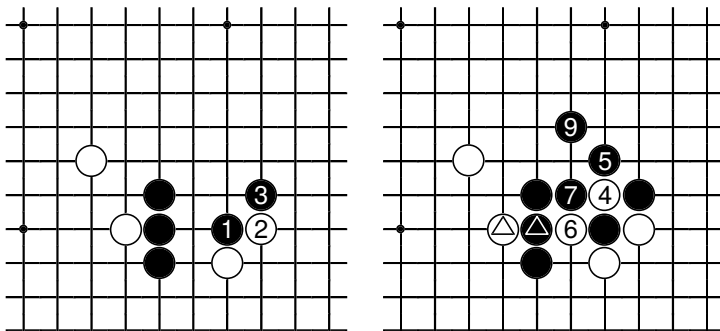
(Left) Black plays the firm bamboo joint shape with 5, to prevent White connecting along the edge. **(Right)** Without the White stone on the side, Black might instead play this way. The cutting point at A remains, Black 5 works to take one liberty off White's two-stone chain, and Black has the chance of making an eye with a play at B.



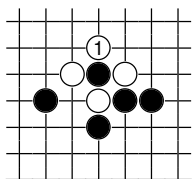
(Left) Because White has bad shape Black 1 here is hard to answer. White cannot prosper in this fight. Therefore the cutting point needs immediate defence. **(Right)** In some cases Black can even set up an obstinate *ko*.



In contrast consider what may happen if Black plays 3 as here. If White continues by sacrificing one stone (**right**), Black's result is a failure from the point of view of shape – the marked Black stone is doing little work, it is much too close to a Black group that is now strong. Efficiency is a major factor in go.



In other circumstances Black may choose a different plan (**left**). The intention is to sacrifice one stone (**right**). In that diagram White 8 connects below 4. After Black 9 the Black group has developed good shape in the centre; and the earlier exchange of the two marked stones clearly favours Black. The single stone capture with 4 and 6 is a famous shape (see box below).

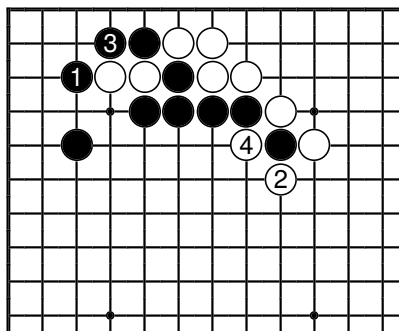


The *ponnuki*

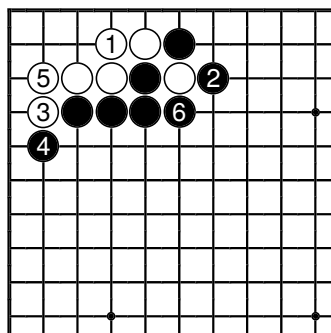
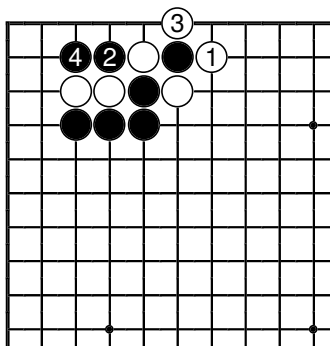
This strong shape, resulting from the capture of a single stone in the centre of the board, is called *ponnuki*. There is a proverb saying **ponnuki is worth thirty points**.

Roughly speaking, that means the three White plays (net – Black has had a stone captured) making up the shape each have value ten points. This is normal for plays in the middle of the game. But if the *ponnuki* is confined, as in the example shown immediately above, its influence is less, and its value also.

When it occurs in the centre of the board, a *ponnuki* may have considerable influence.



This is a common position in which it may be a good exchange for White to make a *ponnuki*, at the cost of many points in the corner. The full value of White 2 and 4 will be seen in the middlegame fighting.



Here, after a 3-3 invasion, it should be better for White to make the capture of the single stone in the left-hand diagram, even though Black then recaptures the corner. The White group formed by the *ponnuki* is quite robust: it has good eye shape already, and White probably doesn't need to add a stone to it immediately.

In the right-hand diagram White takes the corner, but Black has good outside influence. When Black captures with 6 it is a matter of taking off a stone caught in a ladder, before a ladder-breaker is played. It also forms a *ponnuki* shape. Black has a cutting point, but otherwise thick shape useful in fighting.

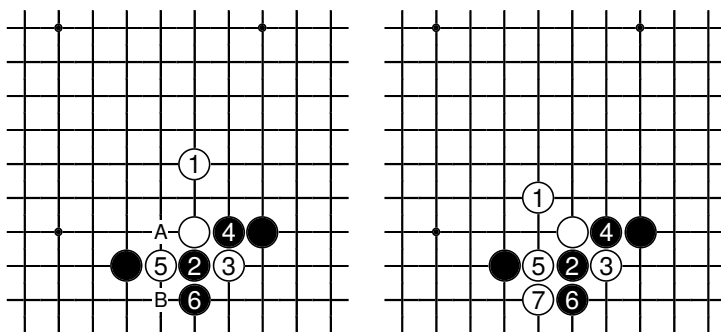
The value of 30 points put on a *ponnuki* by the proverb must be qualified, as we have seen, when it is prevented from exerting full influence; but also when too close to other influence.

Chapter Two

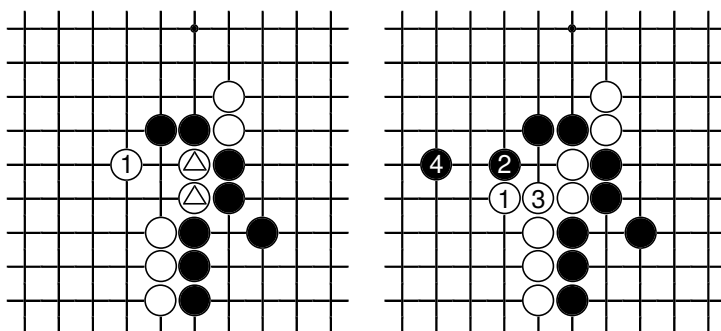
Shape Basics

2.1 Introduction: functions and comparisons

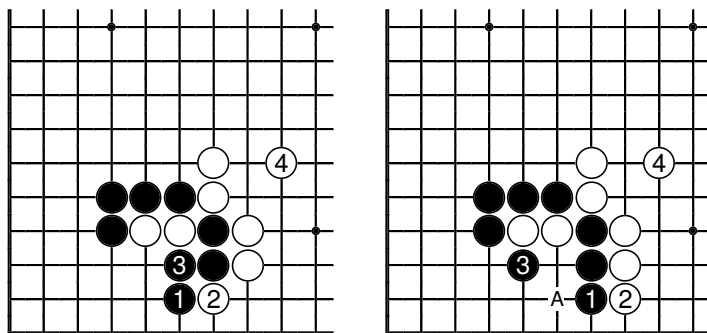
Chapter 1 showed some strong and useful shapes, but you need principles as well. The first steps in understanding shape come not with looking at specific patterns, but with the idea that certain plays work well, where others disappoint.



The one-point jump White 1 in the left-hand diagram doesn't allow White to keep the two Black stones separate: after Black 6 Black will be able to play at A or B. But the diagonal play 1 (**right**) works perfectly.

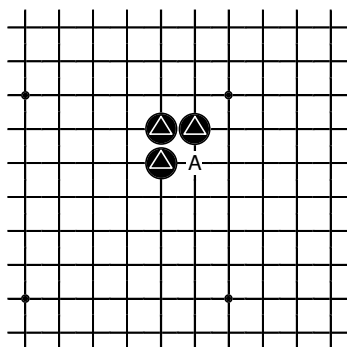


The one-point jump (**left**) is the way to defend the two marked stones, since the diagonal play (**right**) allows Black the snapback threat at 2.



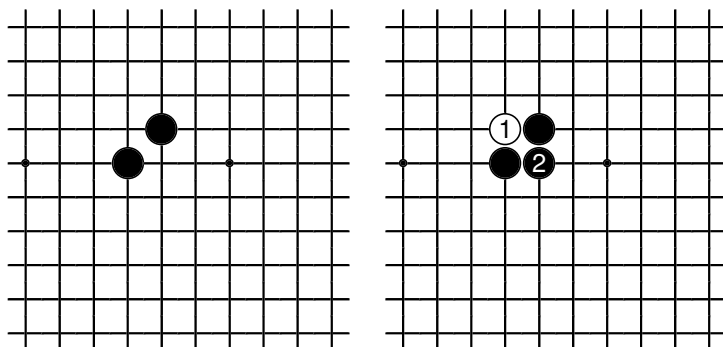
Black should use the diagonal play 1 (**left**) to capture the two White stones. Playing down towards the edge (**right**) is clumsy: Black 3 is needed because of the threat of White A. Now Black cannot capture a white stone on the edge, a big endgame play that also creates a cutting point.

2.2 Empty triangles – every dog has its day

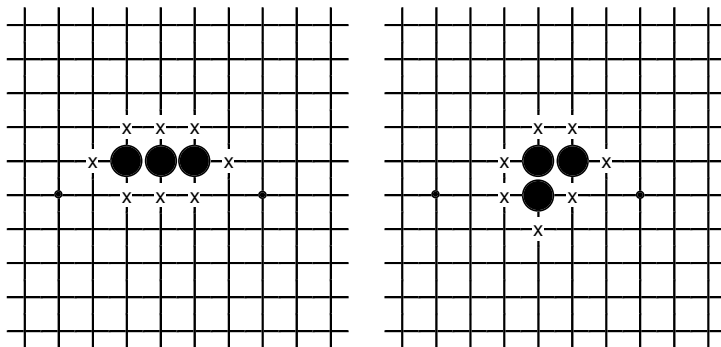


Those who have learned as axiomatic that the *empty triangle* – this marked Black shape, in which the point A isn't occupied by a White stone – is a bad shape, may be surprised later to find that it has some possible advantages. From the point of view of 2.1, that's a matter of seeing how it matches the needs of the position.

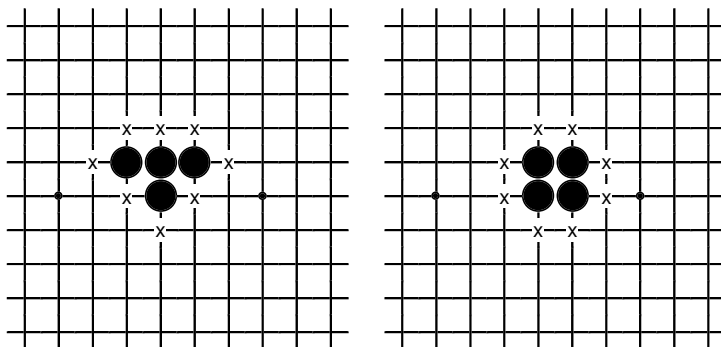
To start with, two reasons that the empty triangle is considered a bad shape.



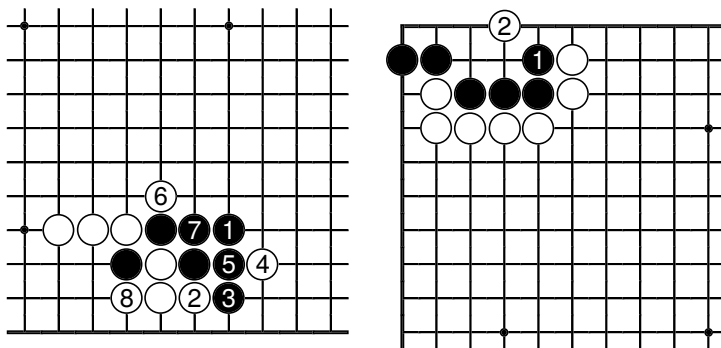
Firstly, it is inefficient. If Black makes the diagonal shape in the left-hand diagram, the two stones are already connected. If White plays 1 in the right-hand diagram Black can connect with 2; and *vice versa*.



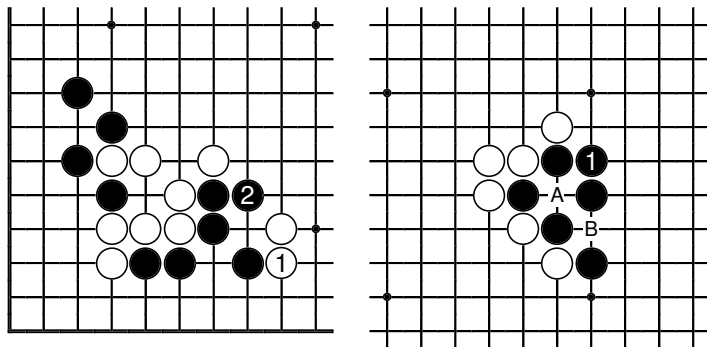
Secondly the shape formed is short of liberties. Three stones in a line (**left**) have an initial eight liberties. In the empty triangle (**right**) they have only seven. This difference may appear to be quite small, but soon makes itself felt in any close fighting.



(**Left**) This shape has two empty triangles in it, and has no extra liberties to show for the fourth stone. (**Right**) The 2x2 block is just as short of liberties, but is also very clearly an inefficient shape that uses stones badly.



Some practical cases. (**Left**) Gross inefficiency: Black should just play 1 at 7. (**Right**) Black 1 is wrong and White 2 kills the corner; Black could live by playing the key point 2 (cf. 4.6).



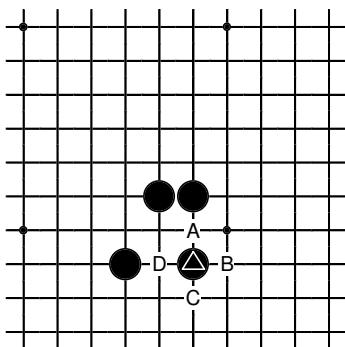
For contrast, two examples of empty triangles as fine shape. **(Left)** Black 2 avoids being shut in. **(Right)** If Black cannot fight the *ko*, this play 1 starts to look good after White A, Black B.

While an empty triangle is bad shape more often than not, it seems impossible to give an exhaustive list of cases where it may qualify as good shape. There are examples on pp. 12, 26, 49, 55, 105, 117, 171, 188, 192 and 195.

2.3 Around the table shape

The table shape from 1.1 was given without much explanation. That is typical of unadorned comments ‘it’s good shape’. This section looks at two ways forward from there.

Adjusting a single stone



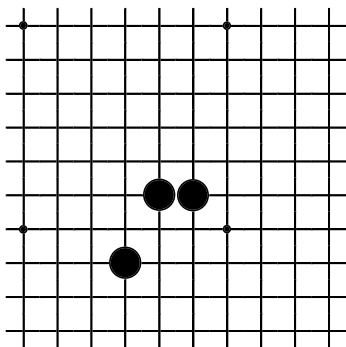
The marked Black stone is most often better placed where it is, rather than at any of A, B, C, D. Why is that? For example A makes an empty triangle shape, not intrinsically a good idea.

The shape made with D is comparable with the bamboo joint. You could say that A and D are too close to the other black stones, so they may be less efficient than the other three plays. On the other hand B and C may turn out to be too far away. If that’s the case then the marked stone achieves a kind of balance.

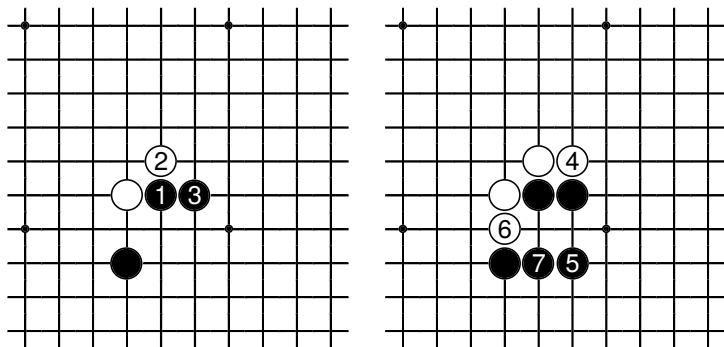
However that's not the whole story. The shape made with C turned out to be good shape when seen in 1.4 and 1.5. It occurs also in the problems, later. The shape made with A is in 3.5L, in a very particular pattern. The shape made with D is very important (for example in Chapter 8). Perhaps only the shape after B is really unusual, and relatively rare as a good shape; when it occurs on the next page it is too loose.

Development and foresight

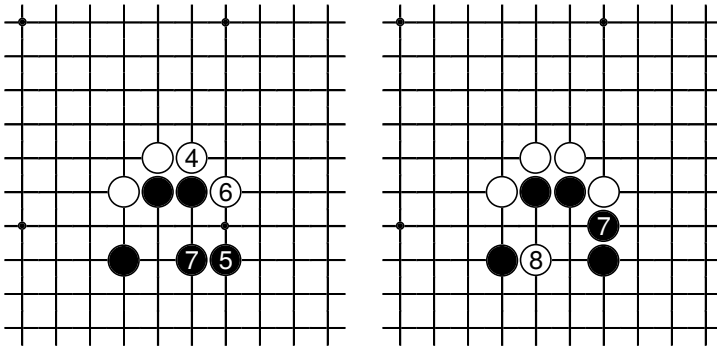
Stones are placed one by one on the board. Any shape more complex than the diagonal or jumps must be put together in full realisation that the opponent may intervene. The first ideas about shape may come from static patterns; but there is the underlying process to worry about, too.



We study this portion of the table shape. It contains three stones, but feels incomplete as it stands. It is expected to occur with some White stones, which might be distributed round it in a number of ways. The discussion of whether the Black stones are properly placed and connected becomes interesting and not too simple-minded.



Now add some White stones, to reduce the level of abstraction, and bring in possible tactical variations. The development in the left-hand diagram is quite normal (more on this in Chapter 5). Black 5 makes a very solid shape (**right**), which is only confirmed when White plays 6 and Black answers at 7 for a bamboo joint.

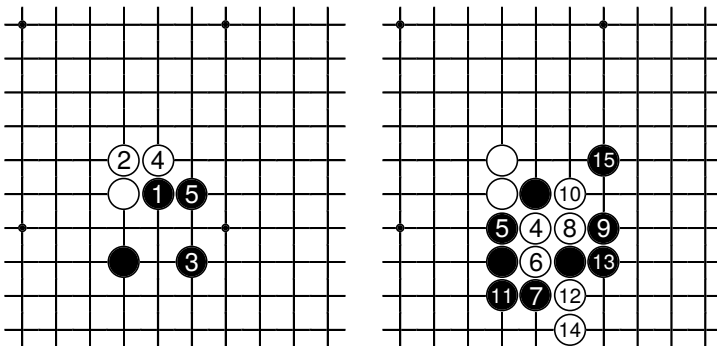


The looser choice of Black 5 here can bring some redundancy or leave future trouble. **(Left)** Here 5 and 7 are too close to each other, and Black's shape is somewhat inefficient. **(Right)** The placement play 8 leaps to the eye. It is a skilful way later for White to cut Black.

Let's ask why, though, this play White 8 is visibly on a vital point. It could be one of two reasons:

- This is the point to complete the double table shape of 1.1.
- This is the way to make an eye-stealing relationship with the white stone played as 6.

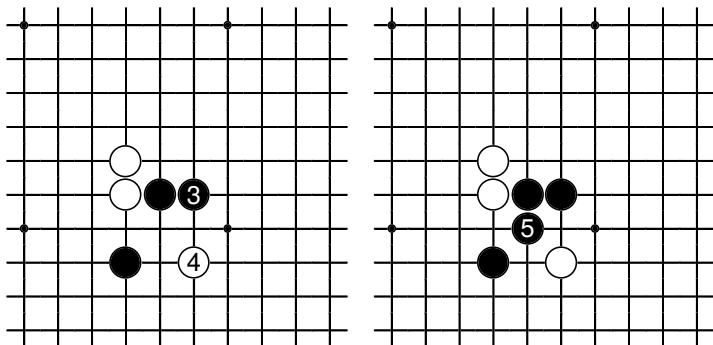
If you think about it, the first reason is related to eye shape too. So here is one way in which shape study isn't so simple: eye shape may be involved in what at first sight is mainly a question of keeping connected.



If White had reacted more passively with 2 here, Black could jump to 3. **(Left)** An efficient result for Black. How dangerous is 3?

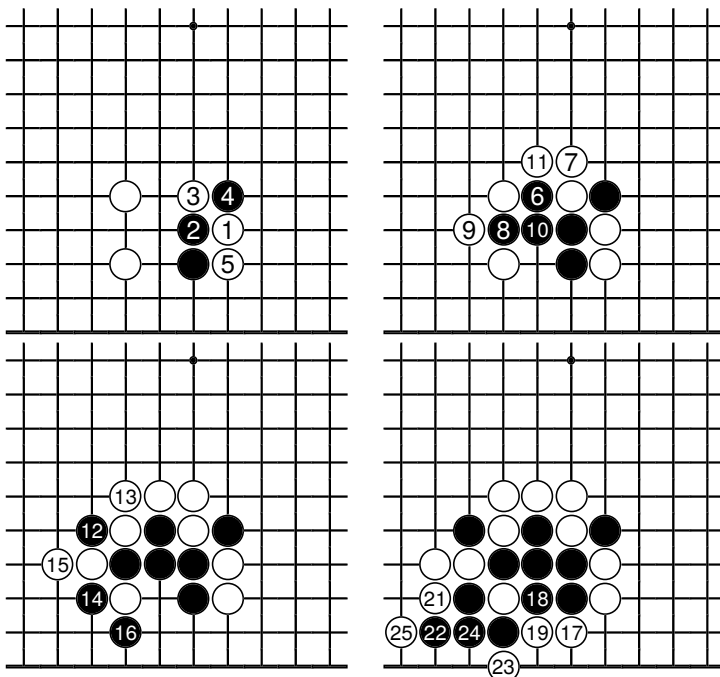
Perhaps White might resist with 4 **(right)**. The meaning of the plays up to 14 isn't hard to understand. Who has the advantage in this fight? After Black 15 it seems that the two black groups will be better developed; it isn't so important that Black has sacrificed one stone.

When there are a number of weak groups in the centre of the board, the balance of dynamic factors in the fighting is the most important thing. One of those factors is the good shape of groups, for running away or eyes.

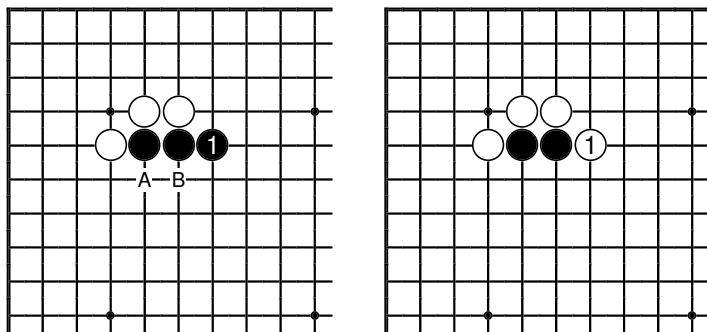


Here 3 is odd: White will take the key point 4. **(Right)** A horrible result. Although Black remains connected, this shape has an empty triangle signalling inefficiency, and also a white stone sitting on a key eye shape point, generally called an *angle* point in relation to Black 5 (4.8, 4.9).

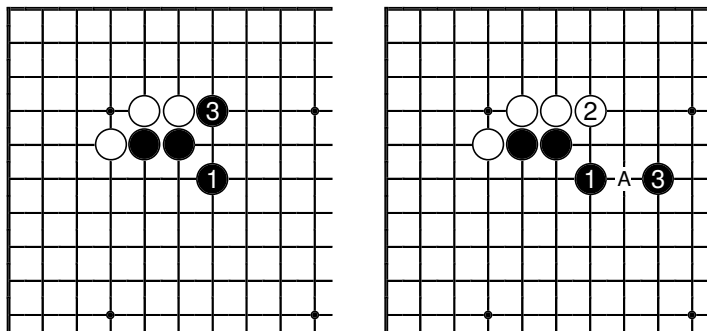
2.4 Fighting: the liberty problem



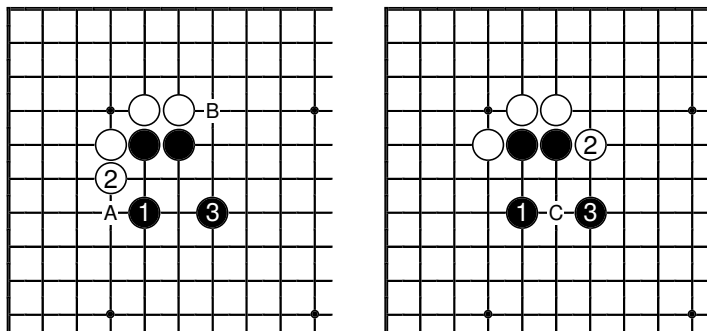
Here's a sad story in four parts. Black tries hard to break out, but simply has too few liberties. As soon as White plays 5, Black's shape is deficient in liberties. The end comes after Black 20 connects, leaving only three liberties. Capturing a single stone with 16 cannot be relied upon to gain liberties, if it only makes a false eye. In fighting even a single liberty matters.



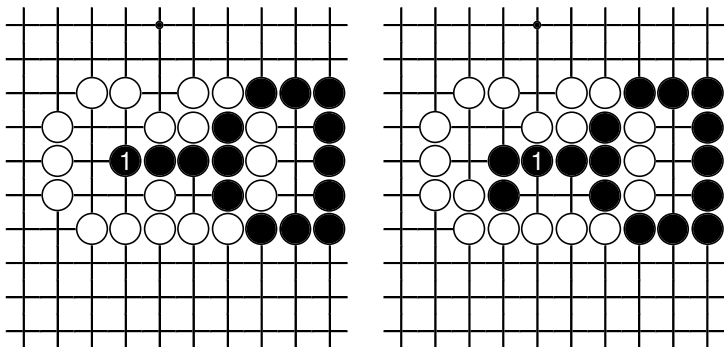
Considering liberties alone, Black 1 is the key point; it gives Black five liberties, while both Black A and Black B make empty triangles, and only four liberties. For that reason White 1 is a good way to attack.



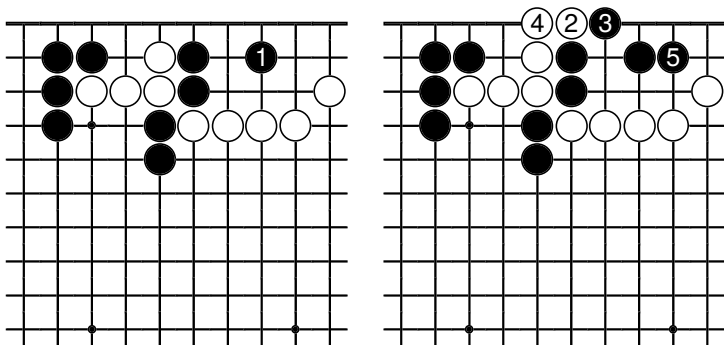
You sometimes give greater priority to adding a second dimension to a group. Black 1 here aims at playing 3 (**left**). But if White answers at 2 (**right**), Black's shape is slack, with a weak point at A. So Black 1 is poor shape.



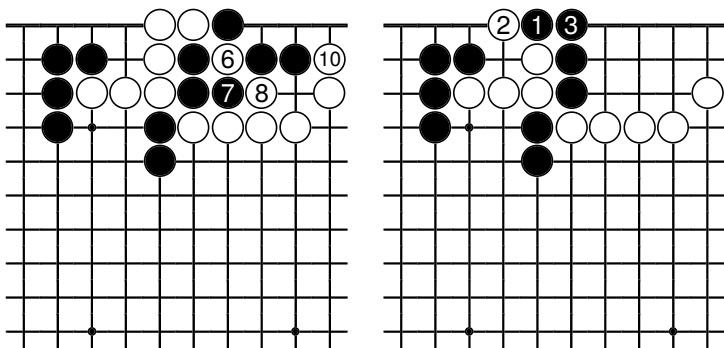
It would be better to jump at 1. If White plays the one of the key points, 2 in the left-hand diagram, Black's table shape with 3 becomes very good, leaving a choice of A or B next. (**Right**) Black 3 is still usually good, but Black will have to watch for the possible weakness at C as the game continues, because White 2 has caused a shortage of liberties. In this case Black might sometimes instead wish to play 3 at C, a bamboo joint. (Cf. 1.3).



The plays shown here are the unique ways for Black to win the capturing races. On the left Black gains two liberties by extending from a chain. (This is a nose play in the sense of 4.3). **(Right)** Black links two chains: other connecting plays, for example the bamboo joint, yield fewer liberties. Empty triangles are *what-you-see-is-what-you-get*, for counting liberties.



In this case Black is behind on a count of liberties. What about Black 1? Can Black gain enough liberties along the edge after 5?



After the throw-in 6 it becomes clear that Black is behind in the race (9 connects at 6). In fact Black should concentrate on reducing White's liberties (**right**). There Black wins because White must connect. Black 1 or 3 is the key point in this capturing race.

Summary on fighting shape

The study of shape isn't about how to avoid fights, but how to enter them only on your own, well-positioned terms.

The key factors in close fighting are liberties, eye shape, and connections. The quickest way to lose a local fight is to reduce your own liberties by clumsy play. The common patterns of liberty shortage are fairly easy to pick up. On the other hand detailed discussion of connections takes up much time in an introduction to shape.

Some principles on liberties in close fighting:

- adding a stone to a chain normally adds one or two liberties, of which your opponent can fill one next turn;
- it may be hard to recover from losing a liberty; so avoid all unmotivated pushing into the opponent;
- false eyes and hanging connections can cause loss of apparent liberties, as in the last example; the same is true of bamboo joints.

At various times in games you will have to worry about:

- whether you can connect (many difficult aspects);
- whether you should connect, and how then to do it;
- whether you can cut, and how best to cut;
- cuts leading to weak groups and running fights;
- peeps, i.e. threats to cut, and how to answer them.

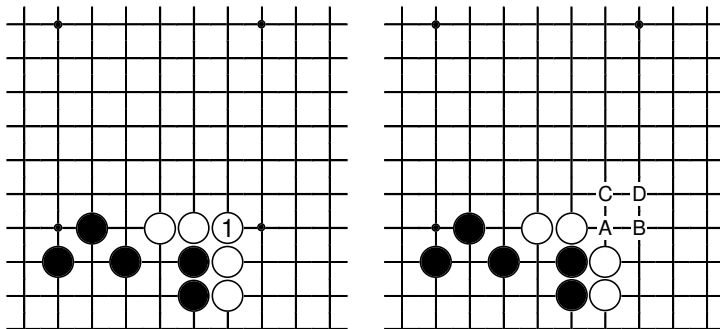
These all may involve shape reasoning. Ambition always stretches out, for extra efficiency; the usual reason for holding back is that you may be cut. (There is also fear of invasion.)

Shape that is capable of making eyes attracts the attention of all experienced players. Some reserve eye shape in groups increases the range of possibilities for fighting in a given part of the board; while taking care of a group that is only just alive may be quite constraining on your other projects.

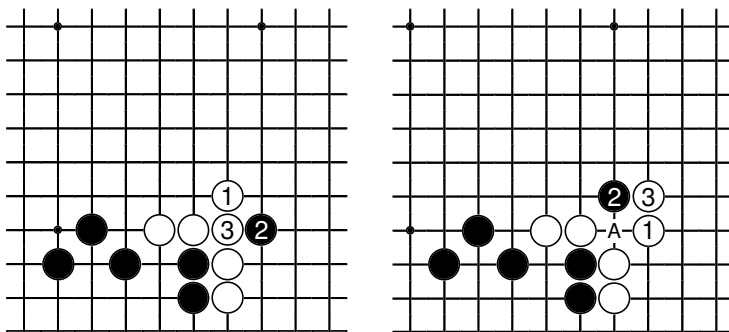
The subject of eye shape is complicated. There are good reasons why whole books are devoted to life-and-death, especially on the sides and in the corners – and in the centre context becomes crucial.

Always pay close attention to context in fights (neighbouring stones, influence, the side of the board); theoretical good shape is a big help, but isn't enough on its own.

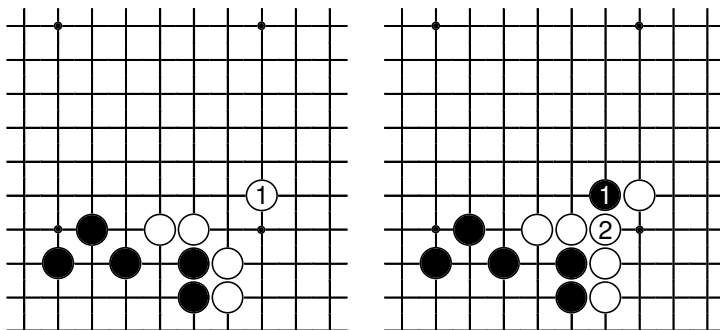
2.5 How to connect



What if anything is wrong with White 1 here? A fundamental question is: how to connect, given a range of possible plays such as A to D. White certainly ought not to get cut round here, but which is the right play? The solid connection A demonstrates no ambition to be efficient, but allows Black nothing at all in the way of later forcing moves.



The hanging connections C (**left**) and B (**right**) both allow Black a peep 2. If White is strong to the right, B may be better; the peep is a waste if it gets swallowed up. After C White has no reason to connect with 3.



The choice D (**left**) is the normal good shape. This indirect connection is supported by two ladders (and here a net, too). The Black forcing move (**right**) is not worth very much: there isn't a good way to follow it up.

Chapter Three

Close Range Play 1

Imagine you move to a large city. To begin with, you go everywhere by car or on foot. After a while you discover how to use public transport, and your life becomes a little easier if less individualistic. It is part of the experience of coming to belong, in your urban environment.

This chapter and the two that follow it are at the heart of our conception of shape. After that, in Chapters 6 to 12, we look at special topics that relate to particular kinds of fight. But first we try to lay down the basic principles of close range play. In order to feel ‘at home’ as a go player, one has to learn gradually the underlying map of the way stones interact. As our metaphor suggests, this may be a slow process depending on changing some habits and conforming more to the usual patterns.

The idea of fighting used in Chapter 2 needs expansion to show how shape really works. In this chapter we concentrate on monochrome (same colour) aspects of development of stones. These are of two kinds:

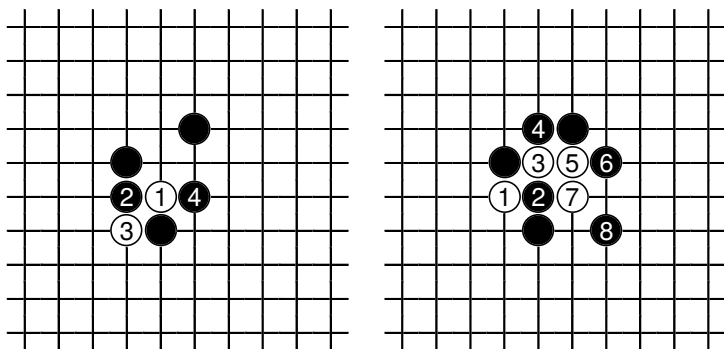
- one-dimensional, i.e. stretch further while remaining connected;
- two-dimensional development, which includes future eye shape but also shape for guarding indirectly against cuts, bends and bulges, and territory-forming shapes.

In Chapter 4 the emphasis is on plays at key points of the opponent’s shape (*hane* plays, the centre of three stones, angle plays as spoilers of eye shape). If you consistently occupy these points, or rather if your opponent allows you to have them, you will gain great advantage in local fighting without having to capture, or even cut. Unless the two players’ knowledge of these vital points is comparable, the game may quickly look like a mismatch with one player’s formations bent back on themselves, and short of liberties and eyes; in short, like a handicap game.

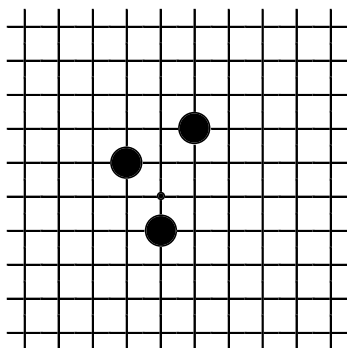
Chapter 5 picks up on shapes after the main types of contact plays. It is normal for contact fights to settle matters quickly, but there are many ways. We emphasise getting off to a good start in such fights.

3.1 Tactical aspects of connections

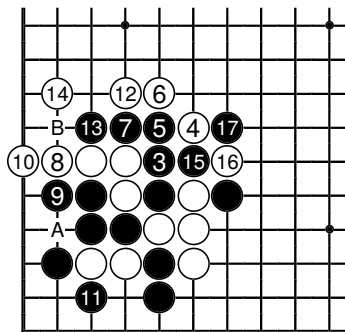
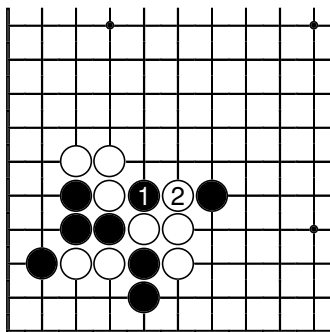
This chapter goes into problems of developing your stones, taking into account short range effects. When it comes to ladders, loose ladders or nets, ‘close range’ has to be taken with a pinch of salt.



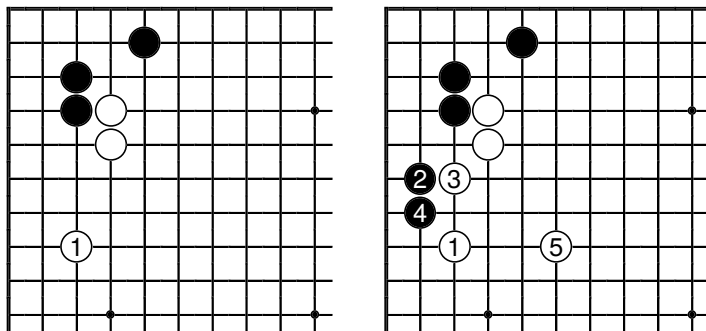
In these simple cases, attempted cutting plays fail to basic tactics.



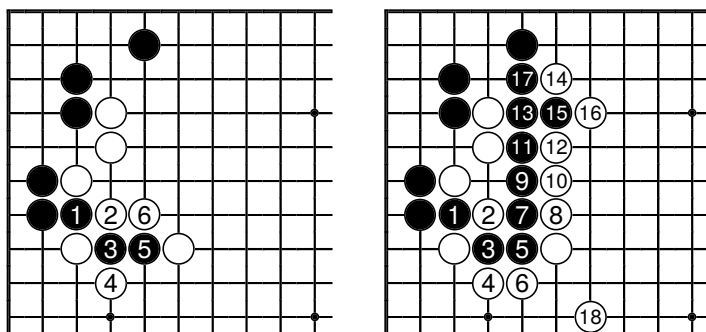
If you look, this shape is made up of two pairs of stones a knight's move apart. A single knight's move may be cut, depending on a ladder. The examples above show that the third stone in the formation is well placed for these ladders (which become a very short ladder, and a net). There are further cuts to try, of course.



Sometimes much deeper reading is required. In this example, does Black 1 work? **(Right)** There are two important variations to take into account. Black must answer White 8 at 9, not A or else White can capture in a loose ladder with 12 and B. White 10 requires a response since it threatens White A. In the end Black survives and White is indeed cut.



An example of the sort of idea professionals use to ‘work their stones harder’, to push for efficiency. This is another possible 3-3 variation, building in a different direction from sequences shown in the Introduction (White 1 jumps the other way, along the top side, in the main variation). White here has central influence and a chance to control the left-hand side. But how sound a play is 5?

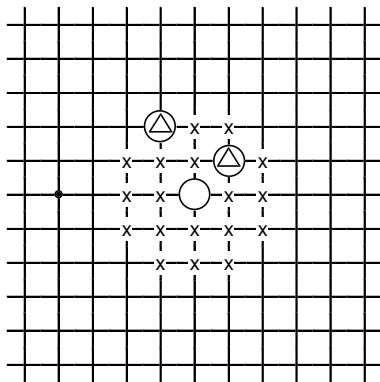


There is a ladder (**left**) that White might use if Black tries to cut. White’s play, however, doesn’t strictly depend on any ladder. If Black cuts White has in mind the possible sacrifice sequence (**right**), if the ladder is bad. So much central strength for White would completely change the middlegame. This is a one-way street; it is natural for Black to push out with 15 to leave cutting points for later. Of course Black 17 can’t be omitted, and then White makes good shape with 18.

Being able to see such sequences makes for much more flexibility; for example White needn’t worry so much about a ladder-breaker. They may involve ladders, loose ladders, nets and netting plays that turn into squeezes. In that sense fundamental capture tactics interlock with with the fundamental shape question of 2.5, “how do I connect?”

The next section looks at a complex example of this kind.

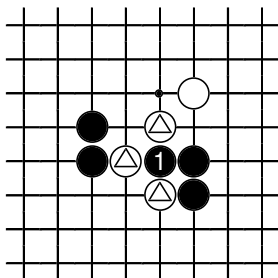
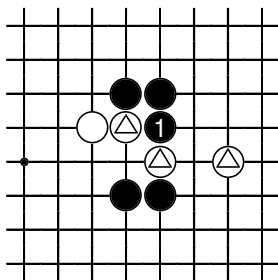
3.4 Compound shapes



Imagine a single stone sitting in the middle of the board somewhere. The most important friendly and enemy stones as far as it is concerned are those as close as the two marked ones, or on the other 'x' points.

What we call a *compound shape* is a shape of three stones of the same colour, not containing any solid connection, that can be made by adding two plays such as the marked ones to the central stone. All 15 types are listed in the next section, to refer to at need.

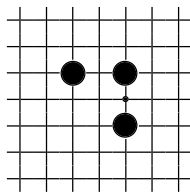
Rote learning disasters



There are some elementary tactical hazards involved in learning compound shapes as connected or not.

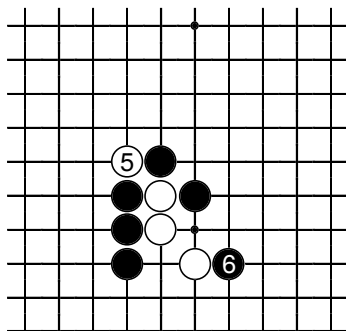
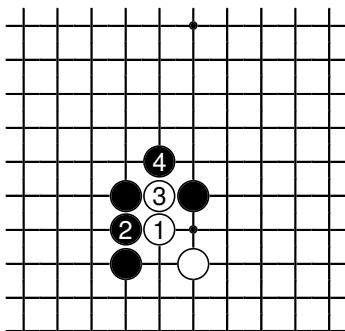
In these examples, an underlying White shape (the marked stones) is properly connected, unless Black is strong nearby. White presumably made them when that was true, and simply failed to check that later developments hadn't changed the tactical position.

In fact the plays Black 1, when put in place, clearly do cut White. This sort of mistake by White is quite common.

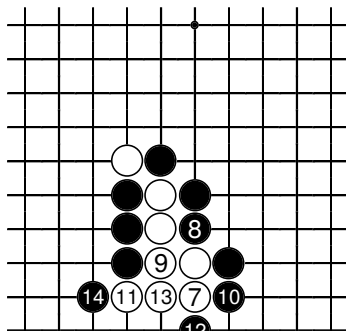
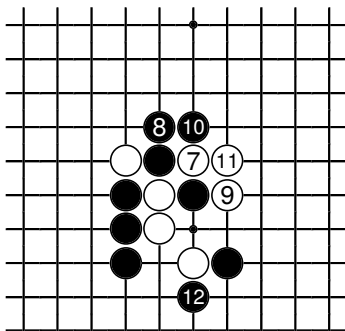


B: The big bend

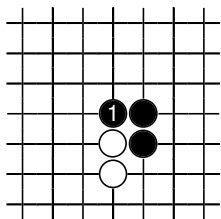
The other way to combine two one-point jumps. It is very often seen with a White stone at the fourth corner of the square.



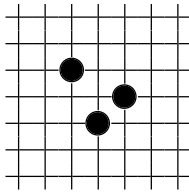
The obvious flaw in this shape is the double peeping play White 1. When this position occurs on the side of the board, Black has the tactical resource at 6. White then may find it impossible to gain from the cut at 5.



(Left) If White moves out in the centre with 7 and 9, Black can bridge under with 12. Normally this fight will go well for Black. **(Right)** Outright resistance with White 7 in this diagram is a simple failure.

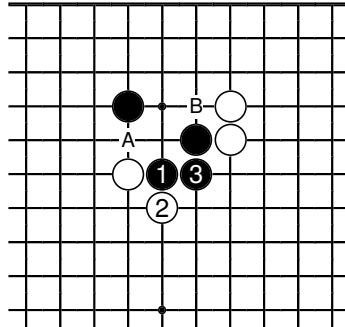
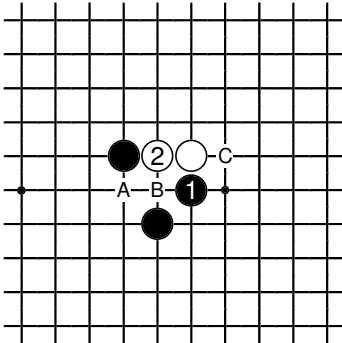


The bend itself, such as Black 1 in this diagram, is a basic shape learned early in everyone's go career. It can make a large change both in territorial terms, and in the balance of influence.

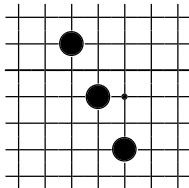


L: The odd diagonal

A special purpose shape. It tends to concede something in the way of key points, so requires a local justification.

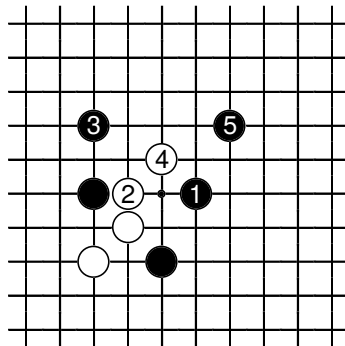
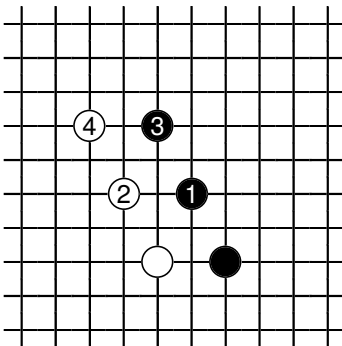


The reason for the name is shown in the left-hand diagram. After White 2 none of A, B, or C is a perfect shape for Black. **(Right)** A very common example. Black emerges into the centre with the *arrowhead* shape 3. If White 2 is at A instead, Black B is good.



M: Attack with the knight's move

This is the classic attacking pattern, in cases when the opponent has no time to cut through.



There are in fact two ways of leaning in such an attack. **(Left)** Black builds a framework to the right by direct pressure. **(Right)** Black may appear to be falling back, but is actually attacking by keeping ahead of White (cf. 4.9). Rapid attacks like these naturally leave behind some weaknesses for Black.

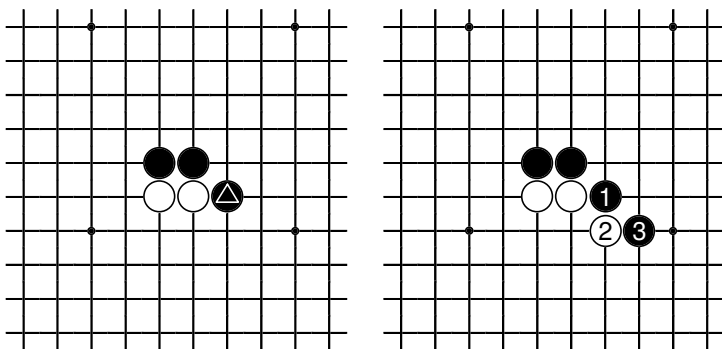
Part Two

Principles of Engagement

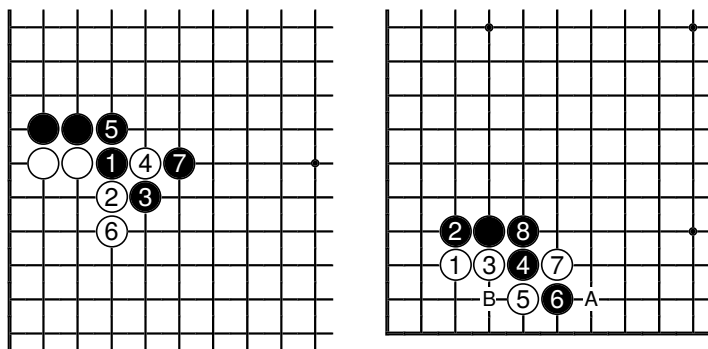
Chapter Four

Starting from hane

4.1 Play *hane* at the head of two stones

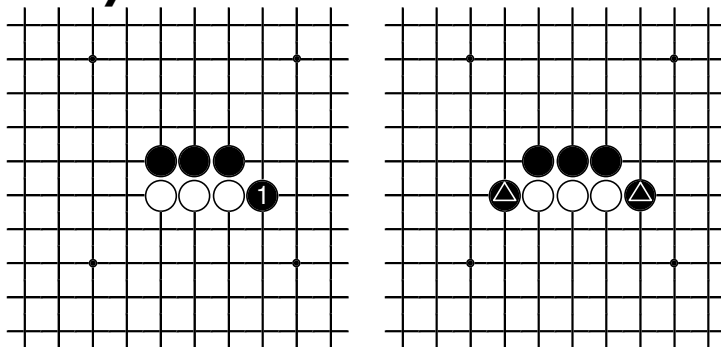


The Black stone marked with a triangle is an example of the *hane* ('hah-nay') play, round the end of your opponent's stones, leaving a cutting point. **(Right)** Black 1 is *hane* at the head of two White stones, and Black 3 is a second *hane*. When it works, this double *hane* is a powerful way to play.

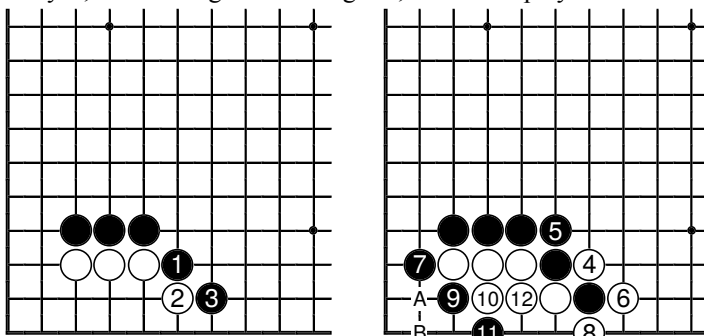


(Left) White must defend at 6, so Black succeeds, bending White back through 90 degrees. **(Right)** After a 3-3 invasion, Black may play double *hane* with 4 and 6, since Black B retakes the corner if White now plays A.

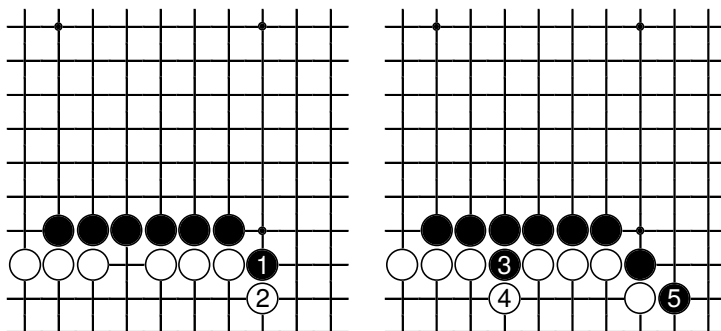
4.2 Play *hane* at the head of three stones



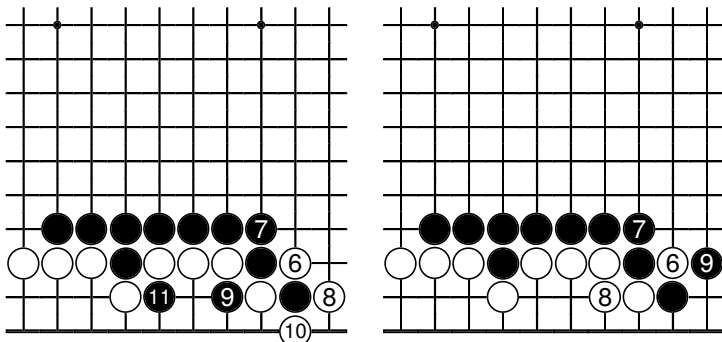
The *hane* play Black 1 also works to put the White stones into bad shape, especially if, as in the right-hand diagram, Black can play at both ends.



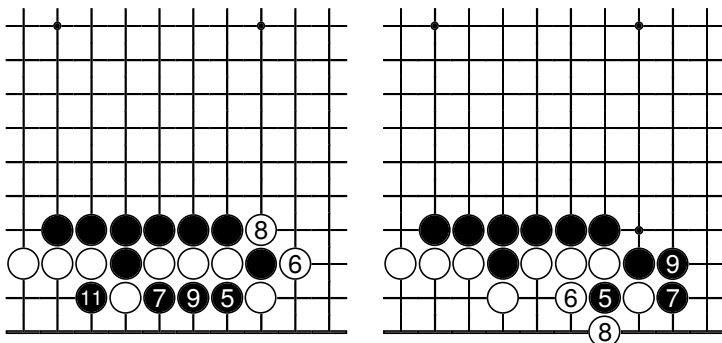
Let's look at the effect of the double *hane* play in the left-hand diagram. While less forceful than the example in 4.1, it still makes White experience some discomfort. **(Right)** White can capture with 4 and 6, but 7 infiltrates the corner. After 12, Black leaves this area for later. White A may be met with Black B for a *ko* fight, but normally Black would play *atari* the other way, sacrificing without serious regrets.



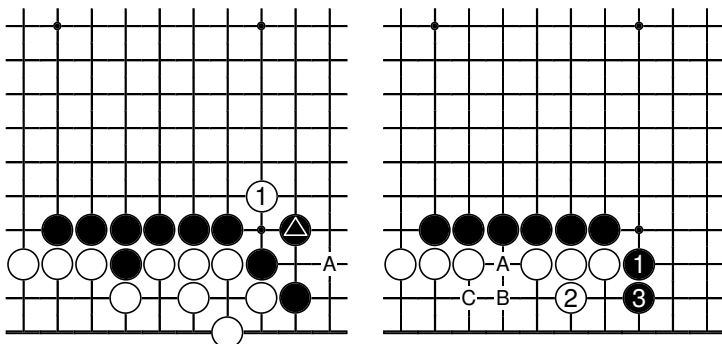
Here is a typical use of *hane* to seal off the lower edge. Up to 5 Black sets White a problem.



(Left) It will usually be out of the question for White to go down this road, losing three stones and being cut. **(Right)** White can choose this way of sacrificing 6, depending on the usefulness of a ladder-breaker.

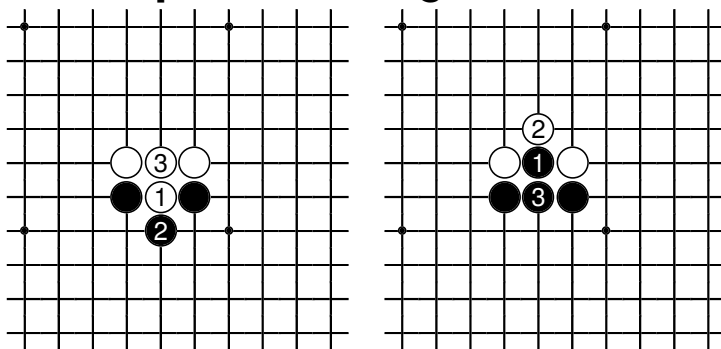


(Left) If the ladder is bad or awkward for Black, there is another play to try. Black 5 here is sharp: White 6 to resist (White 10 connects) will normally look poor after Black 11. **(Right)** Therefore Black may succeed in sealing off the edge with 7, at the cost of White's improved eye shape.

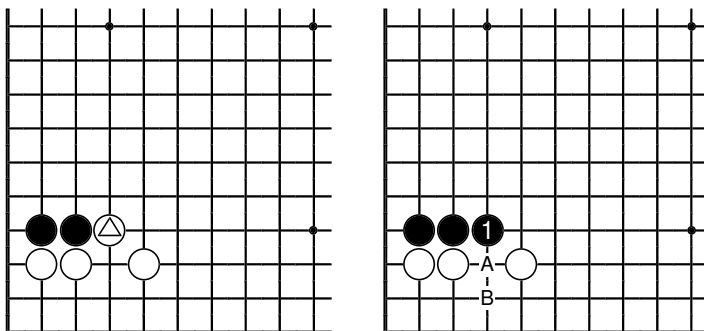


(Left) Black 9 in the previous diagram is generally preferred to a hanging connection, to minimise White's peeping plays such as 1 and A here. **(Right)** Sometimes White may play 2 this way, an empty triangle but not so bad, to take *sente* and guard against the cutting sequence Black A, White B, Black C on the other side.

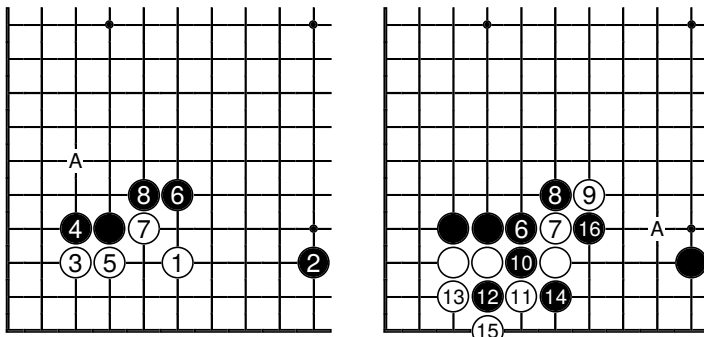
4.4 Don't permit the bulge



This formation, which may occur anywhere on the board, is explosive in shape terms. The plays 1 make a net difference of four cutting points.

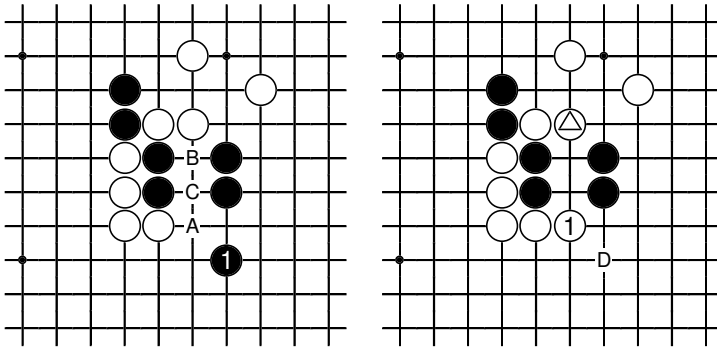


The bulge of 3.5G, that is, the marked White stone, is a key point in many shapes. If Black 1 is played there, Black A, White B, will saddle White with two dangerous cutting points. Compare with the previous diagrams.



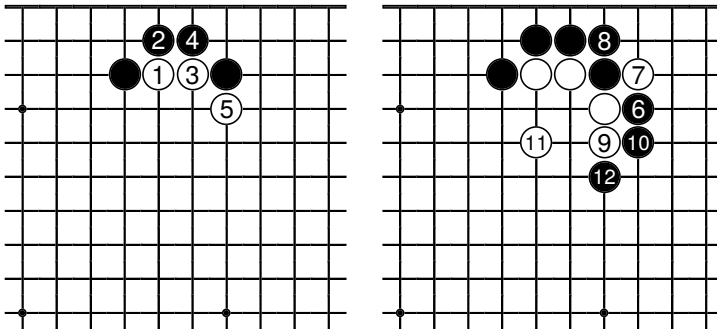
(Left) Black 4 should really be at 5 in this case, but Black 6 is a serious shape error. White 7 sets up a White play at A, to which Black has no very good reply. This is the sort of bulge one shouldn't permit to the opponent.

(Right) Black 6 played at the bulge point guards Black's shape and attacks White's. If White plays 7 and 9, Black sets White a problem up to 16. Black's pincer would now be better at A, which is why Black 4 is wrong.

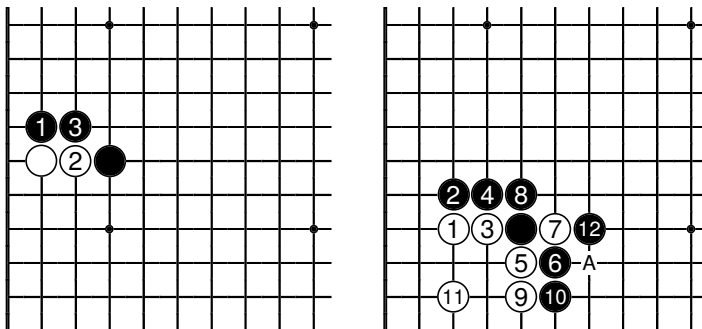


However in this position the bulge point A isn't important for either player. Black should play at 1 as shown; Black A, White B, Black C reveals an empty triangle. **(Right)** White 1 overlaps in effect with the triangle stone, and violates another proverb: *don't peep both sides of a bamboo joint*. It would be better placed at D, since Black has no interest in playing at 1.

4.5 Don't butt towards the centre

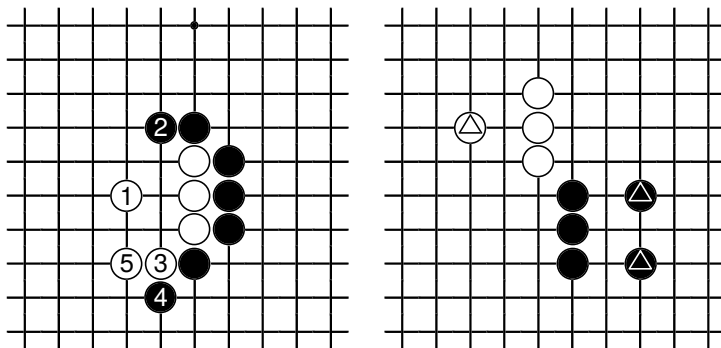


White 3 is a *butting* play, hitting a black stone head on. Up to 12, White's shape is very bad. White has made the same type of mistake twice.

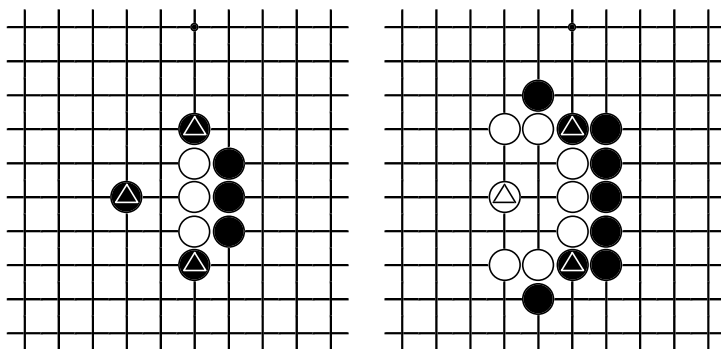


(Left) In general White 2, butting towards the centre, is a pattern of bad shape. **(Right)** White's shape here is poor. After Black's double *hane* with 6, White must defend, rather than play 9 at A.

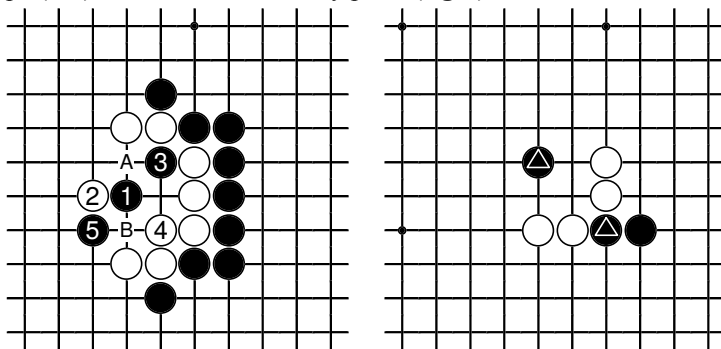
4.6 Play at the centre of three stones



These next sections are pivotal, and integrate ideas from all the first four chapters. White 1 is good shape for moving out. It anticipates the double *hane*, and prepares a table shape. This stone symmetrically placed at the centre of three stones defends a vital point. **(Right)** The single marked White stone is also more efficient than the two marked Black stones.



The triangles formed by these marked stones are ideal examples of attacking shape **(left)**, and defence at the key point **(right)**.



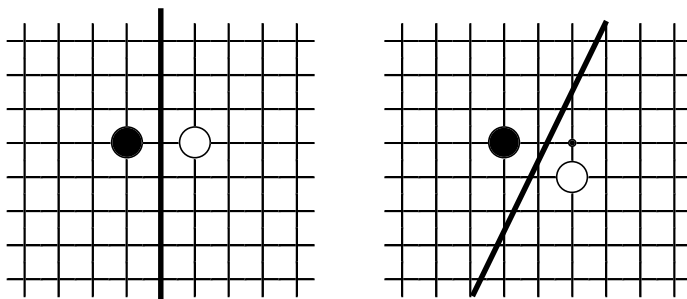
(Left) Black takes the key point, and White's eye shape is gone (White A, Black B leaves a false eye, Black 3 could be omitted). **(Right)** Compare with this basic eye-stealing pattern and the double table (1.1).

Chapter Five

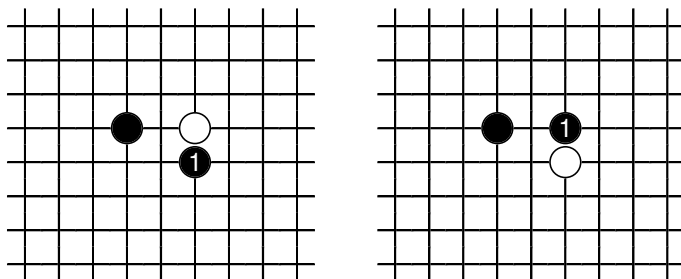
Close Range Play 2

5.1 Approach plays and gain lines

Approach plays (*kakari* in Japanese) are the first elements learned in opening play. They can happen anywhere on the board.

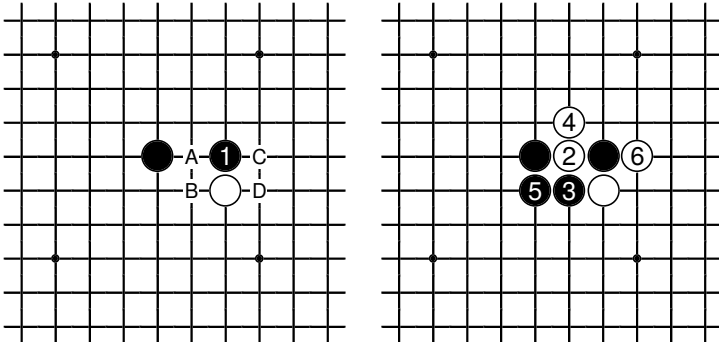


Where they do occur, there is a confrontation, across a *gain line*, marked in these diagrams. One way to get a local advantage is to push your opponent back, relative to these lines.

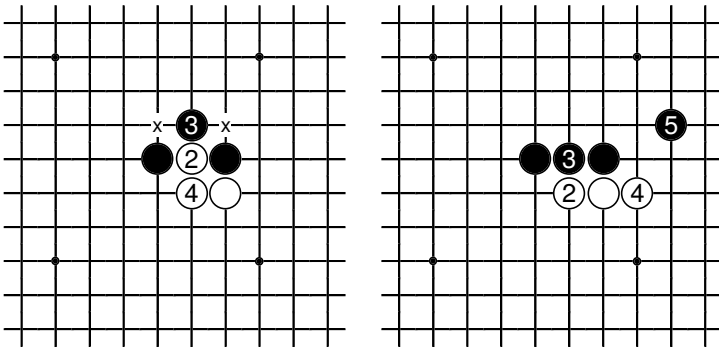


These plays, which are the conventional *supported contact plays*, are then of primary interest. In each case Black 1 tries for the maximum advantage, measured by getting over the gain line, while remaining close enough to the initial Black stone to see some benefit from its proximity. Because of their importance, and the variety of possible outcomes, we devote two pages to each of them.

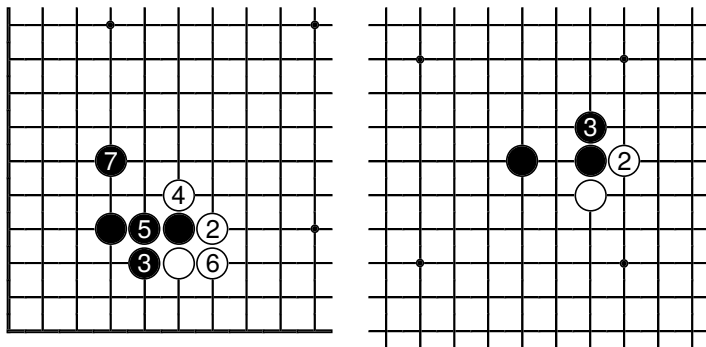
5.3 Answering the attachment on top



White's normal answers are A to D here. (Cf. also 4.4, for a shape to avoid.) White at A can be said to depend on the ladder in the right-hand diagram.

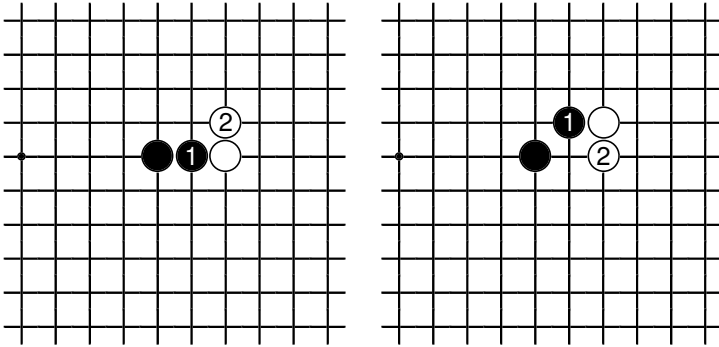


If that ladder is good for White, and Black has to play *atari* on top (left), White is doing well because of the cutting points marked 'x'. (Right) White at B simplifies the development to a trade of influence.

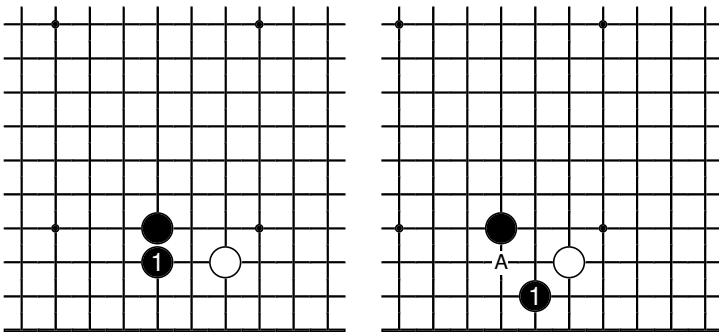


When White answers at C, two standard patterns may occur. (Left) The attach-block shape made by Black 3 is a corner opening, in which Black 7 is important to guard the 'nose' weakness in the corner (see 4.3). (Right) The attach-extend pattern again, which was met in 5.2 in another form.

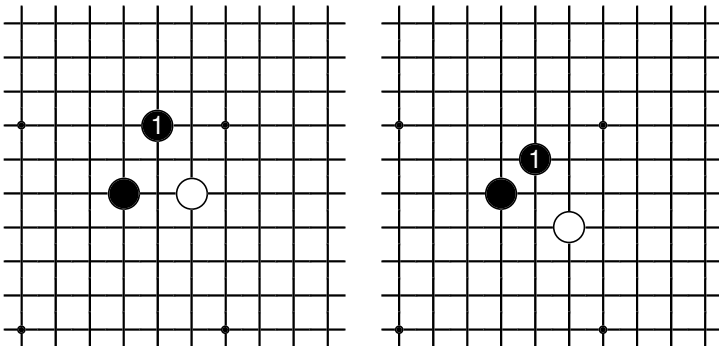
5.4 Restrained shapes



This butting play (**left**) is for special purposes only. It doesn't aim to get over the gain line. It loses out on influence, because Black 1 ends up so close to White. The diagonal attachment (**right**) is still something of a loss, and should be used to attack. In each case Black 1 is a local concession.



The *steel post* (**left**) works well sometimes, when Black can defend territory to the left and also attack to the right. The *footsweep* (**right**) is a harder-working play, but has a weakness at A (see 7.5).

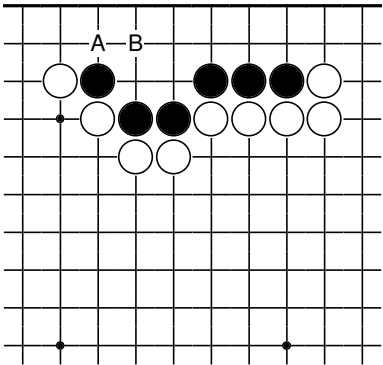
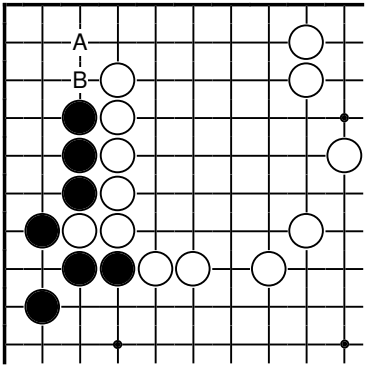


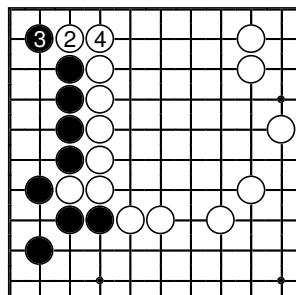
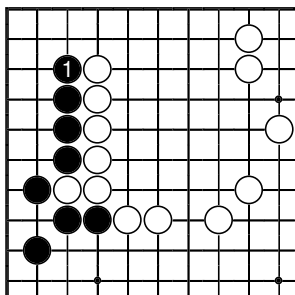
These are constructive plays for influence or central territory. They aim only for a reasonable result, pushing back the opponent. In fact there are real virtues in plays that leave the opponent wondering how to build power.

Problem Set 1

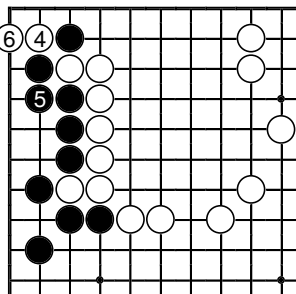
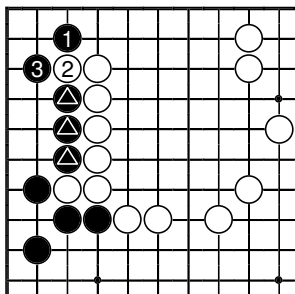
Creating Good Shape

Both problems Black to play at A or B

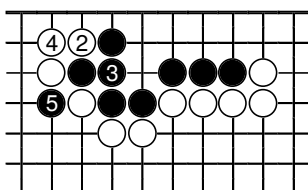
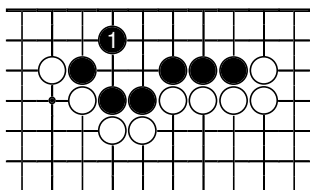




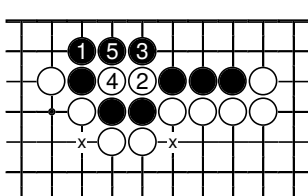
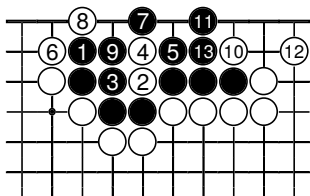
Answer B is correct. Black should simply extend the three-stone chain. Then the strongest play available to White is *hane* at 2, an endgame play worth about 13 points.



If Black jumps into the corner (answer A) White 2 is in the *hane-at-the-head-of-three-stones* position relative to the marked stones. White can capture the corner with 4, threatening 5.

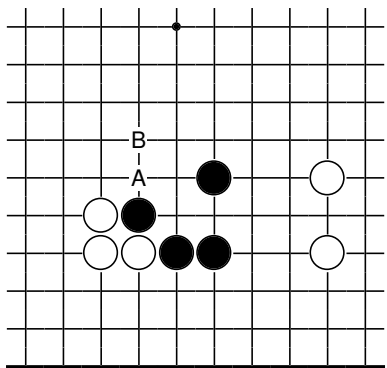
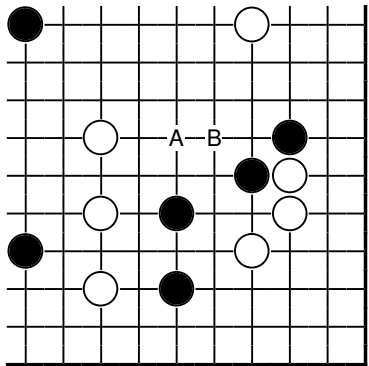


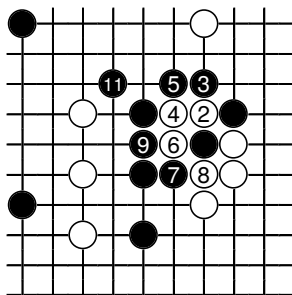
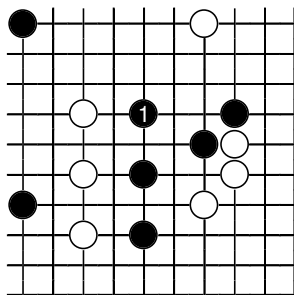
Black should play the hanging connection (answer B). Then Black has a chance to play the cut on the outside at 5, before worrying about the safety of the group.



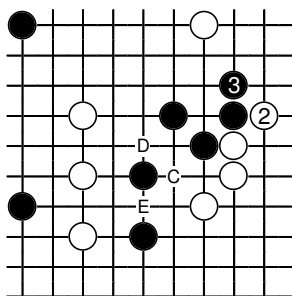
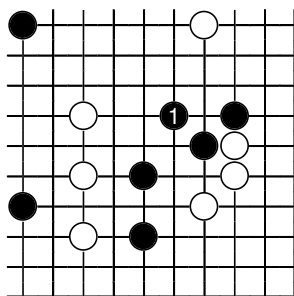
Descending (answer A) is bad shape. If White attacks immediately Black only just survives and White gains influence (**left**). Black's plan of giving up two stones (**right**) is bad, since the cutting points 'x' disappear.

Both problems Black to play at A or B

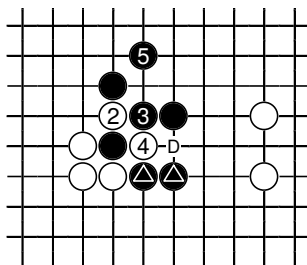
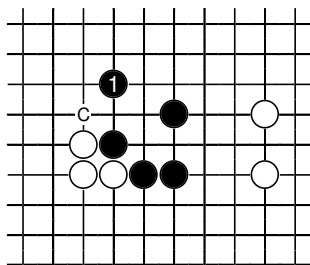




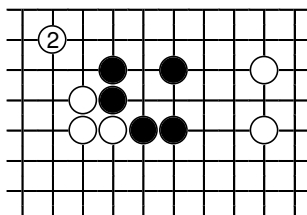
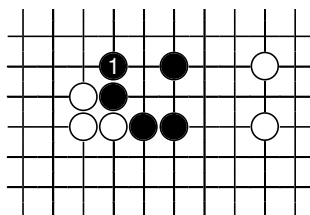
Black should jump to 1 (answer A). This is an example of light shape. If White cuts at 2, independently of any ladder Black can give up one stone and squeeze with 3 and so on.



Answer B, the hanging connection, is a heavy shape. White can play 2, and then wait for a chance to cut with C, Black D, White E.

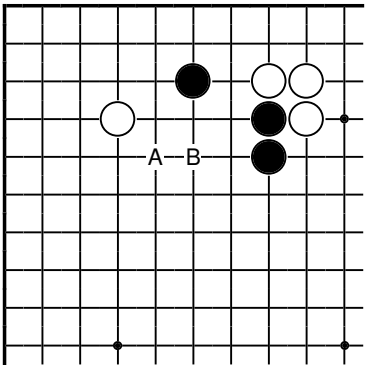
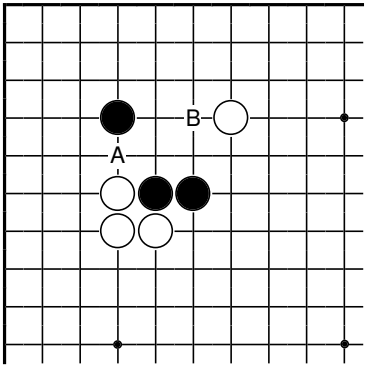


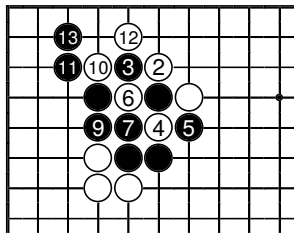
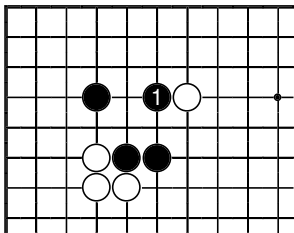
Here again the right answer is B, the one-point jump. Black should play lightly, aiming at the point C, and not worrying about 2 and 4. The weak point left at D isn't so serious.



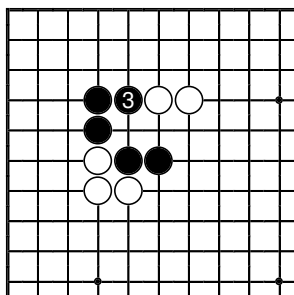
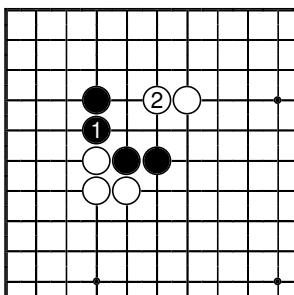
Extending (answer A) is heavy shape. It is easy for White to jump ahead of Black in the running fight.

Both problems Black to play at A or B

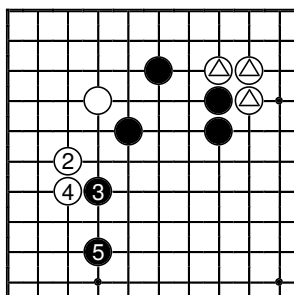
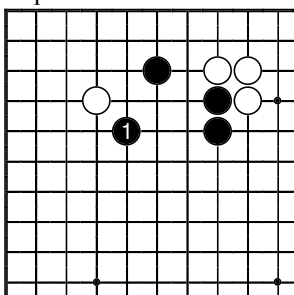




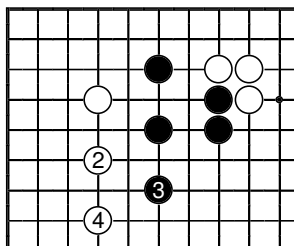
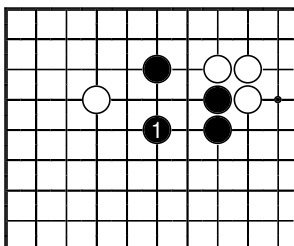
Black should make the table shape (answer B). This is good shape. In this position Black should think first about moving out into the centre. **(Right)** This result would be good for Black (8 connects).



Black A leaves an incomplete shape. **(Left)** White 2 is obvious and best. **(Right)** Then 3 is the proper play for Black, but naturally this isn't very good shape.

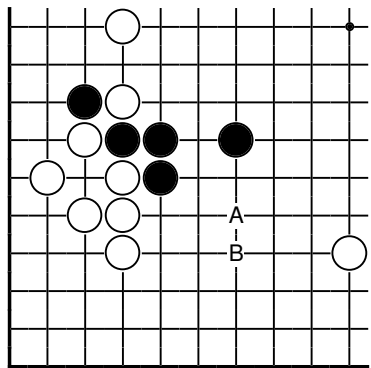
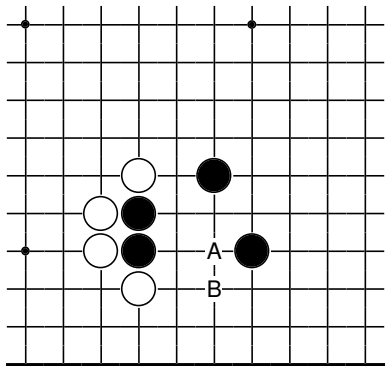


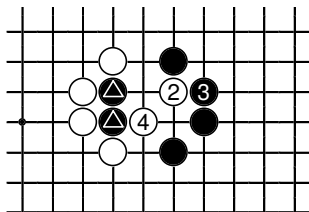
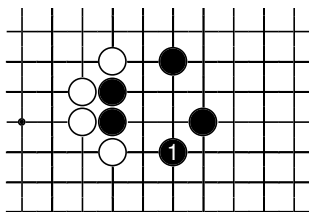
Correct in this case is the knight's move press (answer A). Black moves out quickly, and can soon attack the marked white stones.



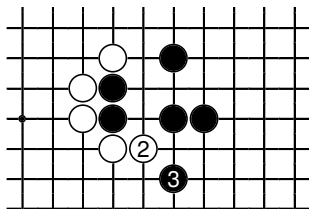
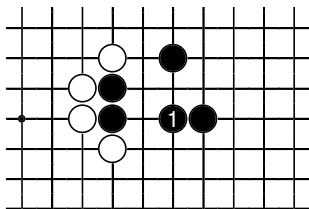
In this case the table shape A is slow. White jumps ahead of Black and takes territory.

Both problems Black to play at A or B

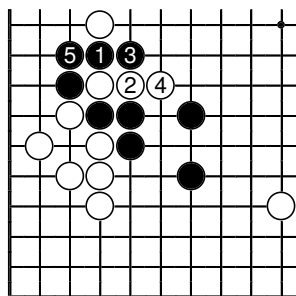
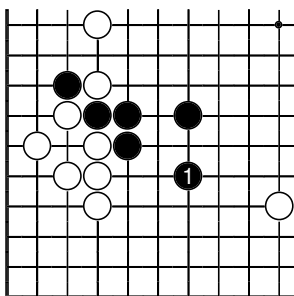




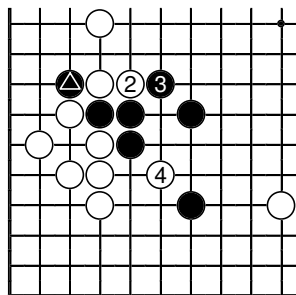
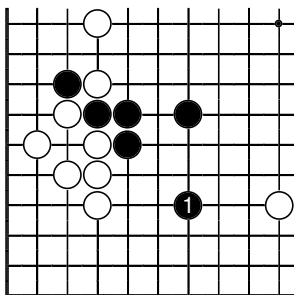
Black should play the diagonal move (answer B). This defends territory. If White spends time taking two black stones with 2 and 4, Black takes profit elsewhere.



The table shape (answer A) is a poor move here. Black's shape is inefficient. White can exchange 2 for 3 and be content.

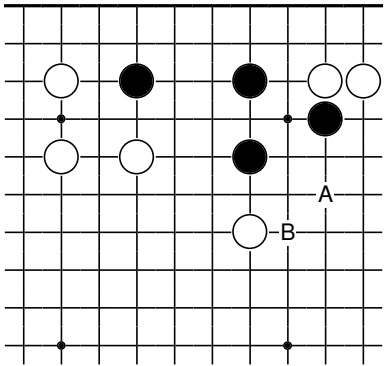
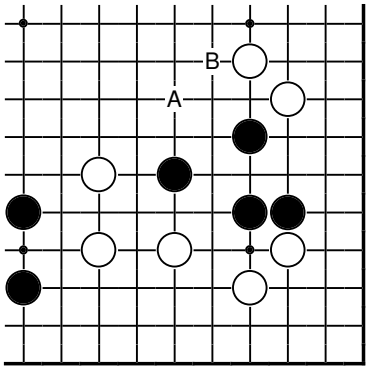


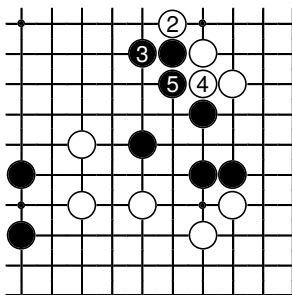
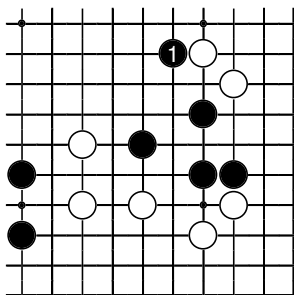
In this problem the table shape (answer A) is best. It prepares for the rescue of the isolated black stone shown in the right-hand diagram.



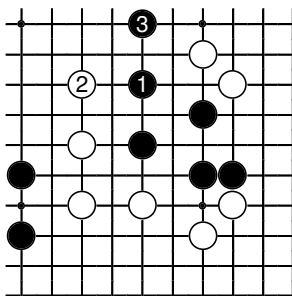
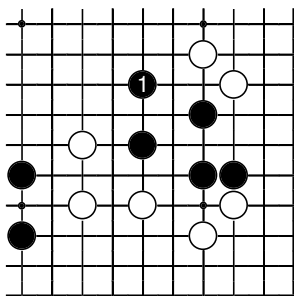
Jumping to B is a loose shape in this case. White pushes up at 2 and then plays 4: Black is in trouble. Therefore Black 3 is bad, though on a bulge point. Since the marked cutting stone also now looks like being swallowed up, it seems that Black has chosen the wrong direction to emphasise.

Both problems Black to play at A or B

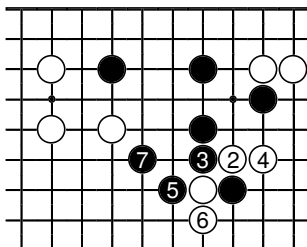
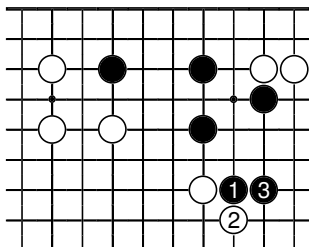




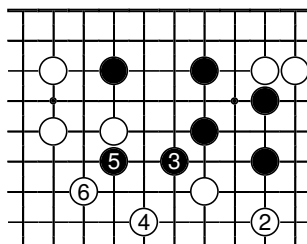
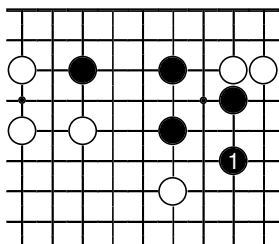
The contact play (answer B) is the way for Black to make good shape here. In the continuation shown Black is out into the centre, and White has a cutting point to handle.



Simply jumping out (answer A) is slow. Black runs out into the centre but is subject to further sustained attack.



The contact play (answer B) is again correct here (the contact play on the other side of the same stone is also good). White 2 (**right**) isn't a problem as Black comes out into the centre with good shape.



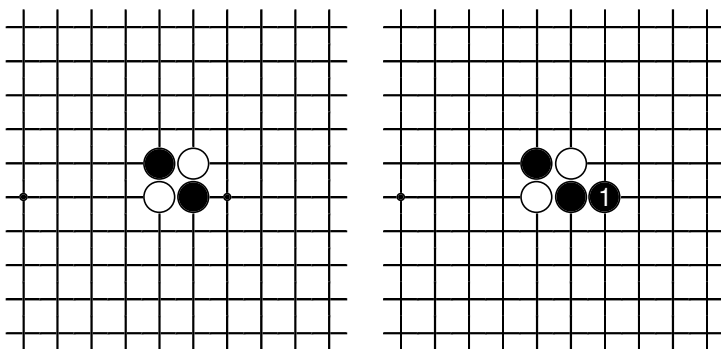
If Black answers at A, it is easy for White to find further attacking plays.

Part Three

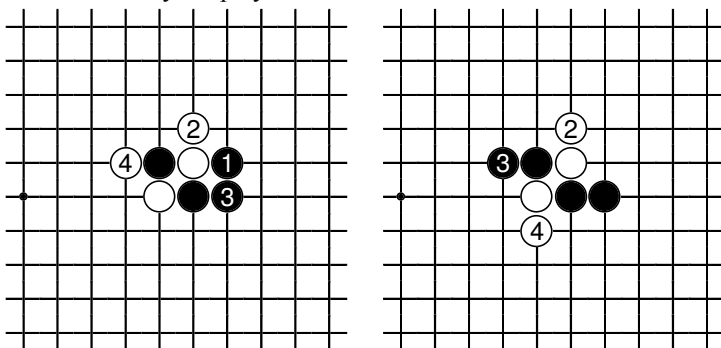
Practical Fighting

Chapter Seven Eight Faces of Cutting

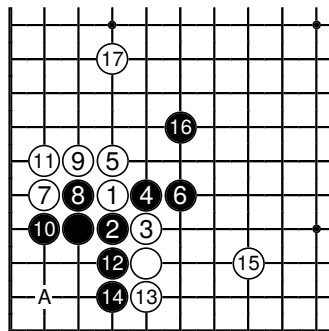
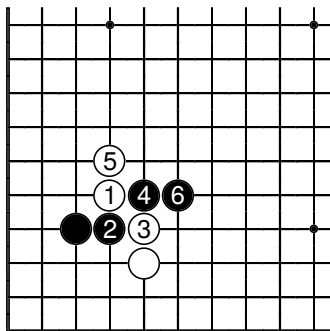
7.1 Windmills to pancakes



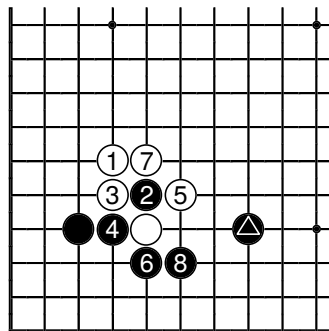
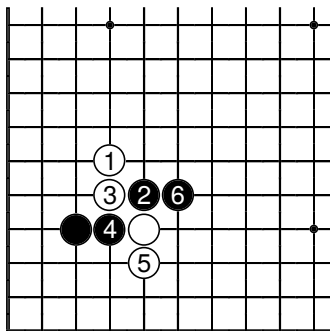
There are a number of fundamental patterns in cross-cut fights. The first is the plain extension Black 1 here. Proverbially, it is better from Black than any of the four ways to play *atari*: ***Cross-cut? Extend!***



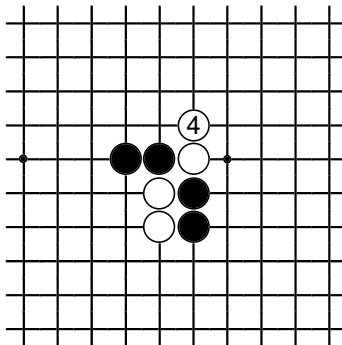
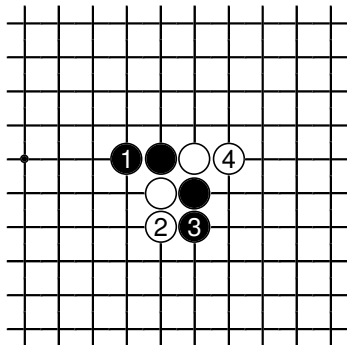
The point is that Black 1 in the left-hand diagram turns out badly, if Black needs 3 also, and White can capture in a ladder. **(Right)** The *windmill*.



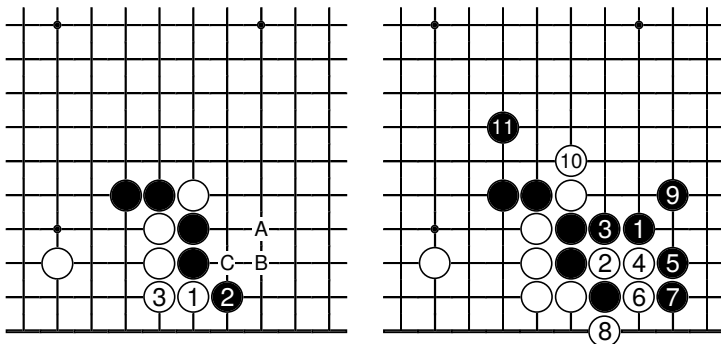
(Left) Black 2 is an overplay, unless Black is already strong locally. Black 6 completes the windmill shape, which gives each of the struggling groups four liberties. **(Right)** White gains the advantage simply by developing on each side. Black 14 leaves the Black corner small and vulnerable (White at A is *ko*), while Black also has a weak group in the centre to worry about.



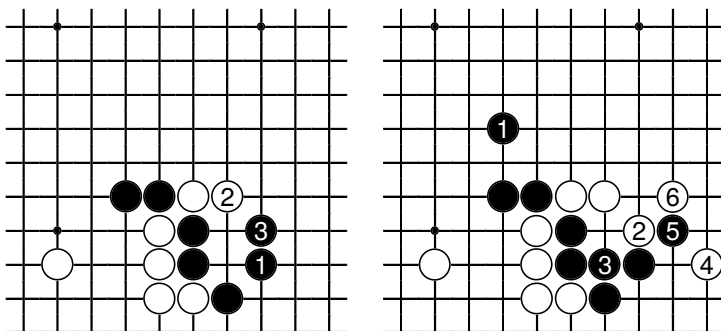
(Left) Black 2 here is also normally an overplay. **(Right)** With the marked pincer stone in place, though, White 1 is questionable. The *ponnuki* plus one stone isn't efficient. If White played 5 at 6, and Black played 6 at 5, the windmill development would be assessed in Black's favour (the lower White group has to work hard for life, the central Black group is relatively strong).



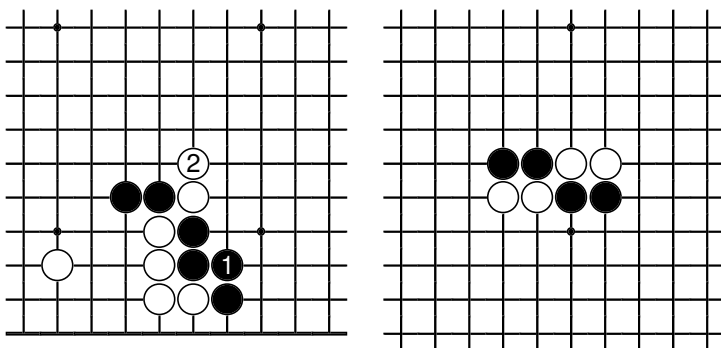
This T-shape **(left)** and asymmetric shape **(right)** also come up, in close fighting. Naturally Black 3 creates a shortage of liberties, for both sides.



Here (**left**) White 1 is clearly better than allowing Black to play at 3. Next should Black play A, B, or C? (**Right**) The idea of Black at A: sacrifice one stone and attack in the centre. White 2 is bad when White is already alive.

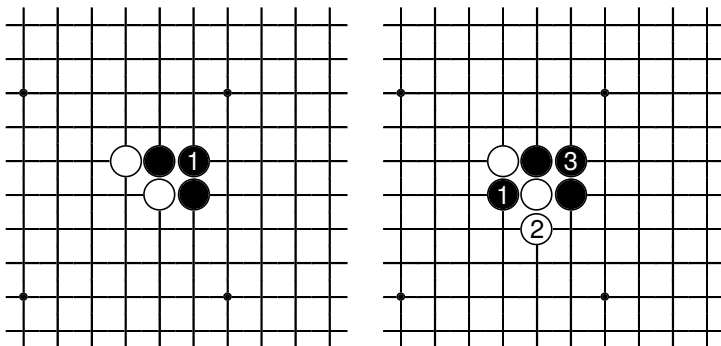


The problem with the hanging connection at B is that Black may need also to play at A, anyway (**left**). If Black omits 3 in the left-hand diagram, White 2 and 4 (**right**) are powerful. (See 9.1 for more on this pattern.)

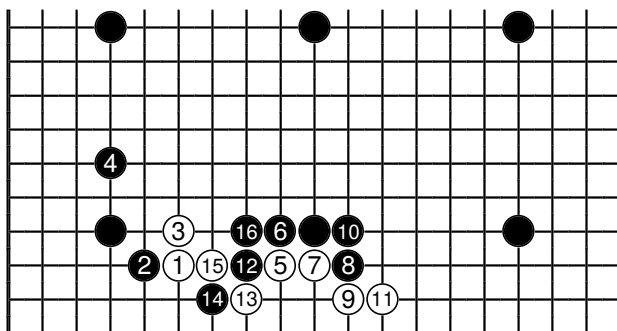


Therefore in some cases (though not this particular one, where White is strong already on the edge), the solid connection C (**left**) may be best. It keeps White weak on both sides, even though it is not intrinsically good shape. (**Right**) We call this extreme example of shortage of liberties a *pancake*. The next page has some cases leading towards it. Naturally enough the fighting can become most difficult.

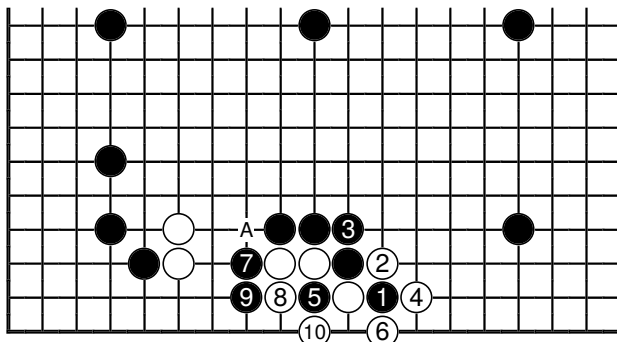
7.4 Staircase connections



Connecting solidly to leave no weakness is usually the prerogative of the attacker. We call the shape in the left-hand diagram a *staircase connection*. **(Right)** The result here is one seen in 7.1. The idea is therefore related to extending after a cross-cut (see 13.2 also): depending on the ladder Black 1 here may turn out to be superfluous, or better played as *atari* at 2.

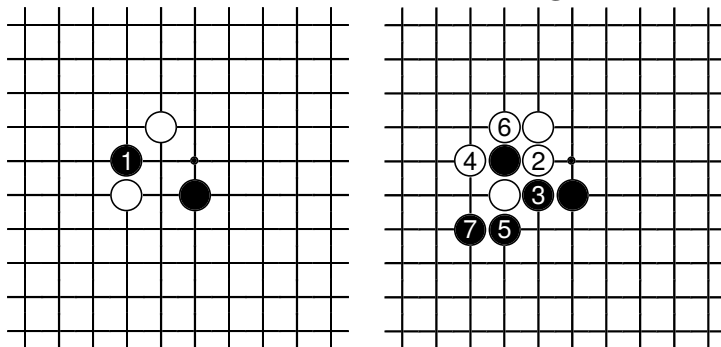


Let's compare Black 10 here, which may seem slow, to double *hane* at 11 (next diagram). In fact White is in a mess after Black 16, and Black solid.

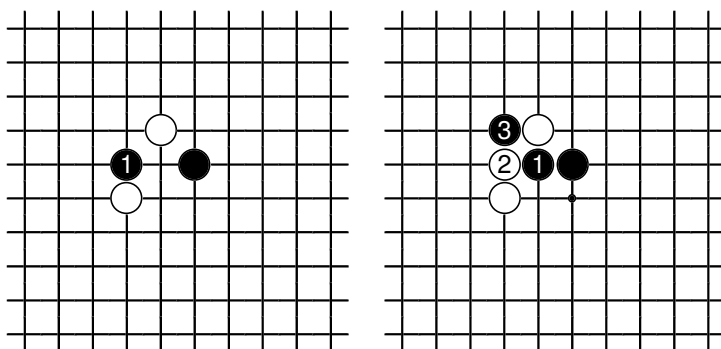


Following 4.1 blindly in this case isn't as good. After 10 White is alive, and has the cut at A to aim at. If Black wants to play double *hane*, a second *hane* play at 7, at the other end of White's two-stone chain, must come first.

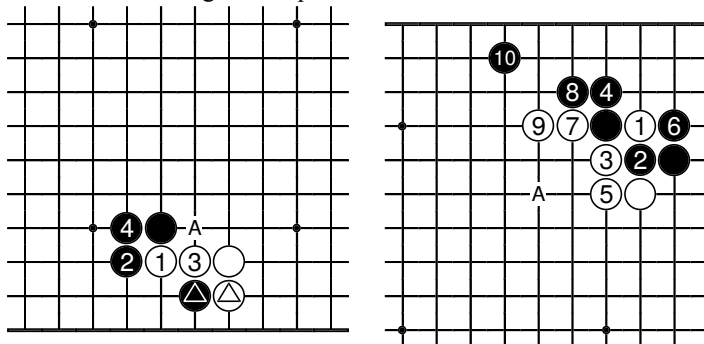
7.5 Strike at the waist of a knight's move



This is a single proverb, with a pair of recommendations. With a choice of how to cut White's knight's move, Black is urged to make the cutting play shown at the 'waist'. This does depend on the ladder (**right**): Black needs to have a good ladder, or be happy to sacrifice the stone played (e.g p.104).

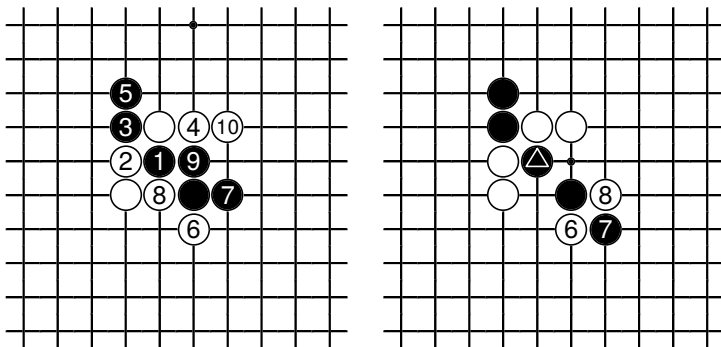


(Left) This is the other cut at a 'waist' (a mistranslated Japanese term that has stuck); again a ladder may matter. **(Right)** In this case however the alternative cut is also good shape.

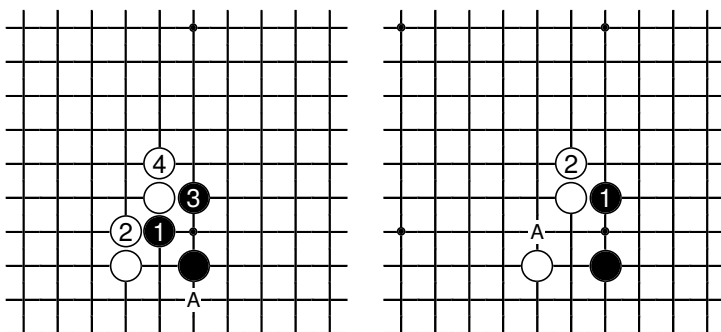


The footsweep, seen in 5.4 and 6.4, is often the occasion for this pattern. **(Left)** In this case Black 2 at 3, White at A is dangerous for Black. **(Right)** A set pattern in the Chinese-style opening. Now A is the key point for both.

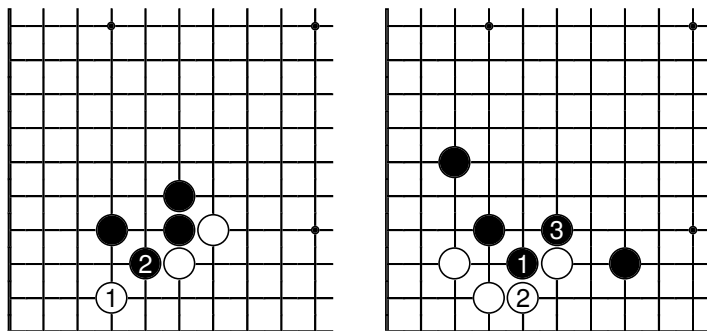
7.6 Pushing into a knight's move



A necessary comment about 7.5 is that White's way of cutting in the left-hand diagram, with 1 and 3, is bad shape. White 8 (**left**) creates an empty triangle. (**Right**) However Black 7 here loses the marked stone after 8.



In general, pushing into the knight's move, without cutting, makes ugly shape. Black 1 (**left**) should be omitted; White A has become a possible good play. (**Right**) Simply playing this way is normally better for Black. There is a chance left of cutting at A, later.

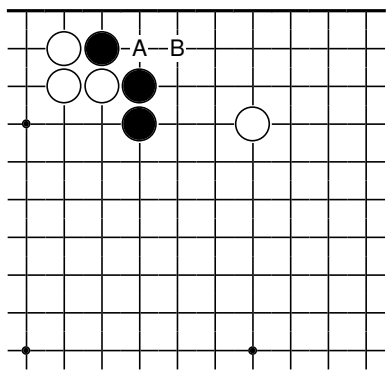
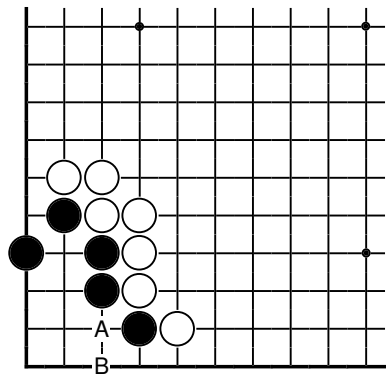


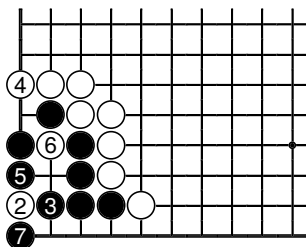
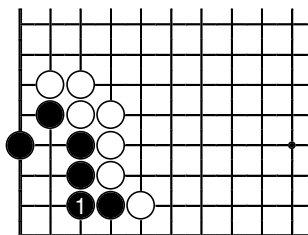
In these cases pushing into the knight's move is acceptable. (**Left**) White 1 is bad shape, and Black 2 is appropriate. White has no good answer. (**Right**) In this opening normally Black plays 1 before 3, fixing the shape here.

Problem Set 2

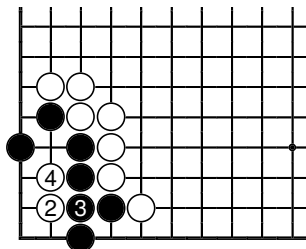
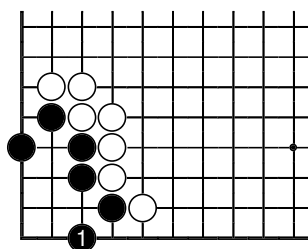
Cutting Points

Both problems Black to play at A or B

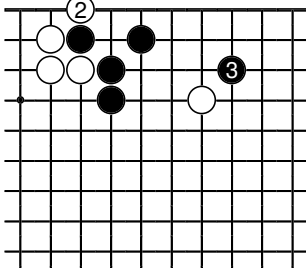
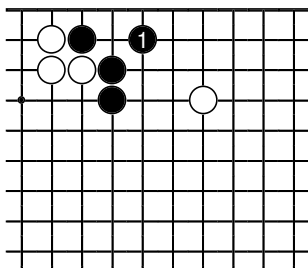




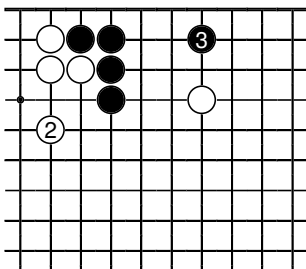
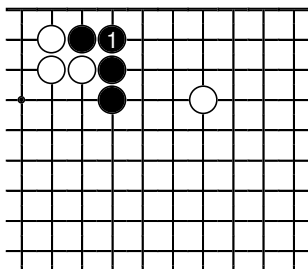
Correct is answer A. This is a common position, and by connecting solidly Black makes immediate life. White's attack in the right-hand diagram is worth playing only in the endgame.



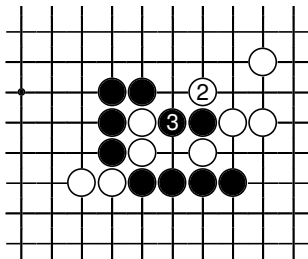
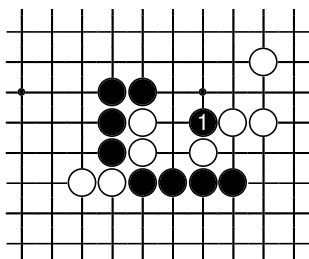
The choice of hanging connection with 1 is bad. When White peeps with 2 the Black corner will be left without two eyes.



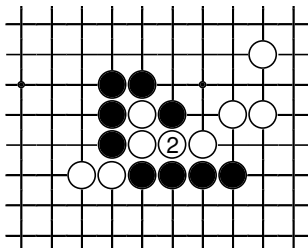
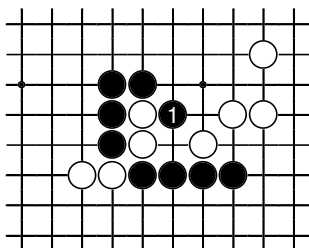
In this case Black should make the hanging connection (answer B) in order to have more outside influence. In the right-hand diagram Black 3 is now possible.



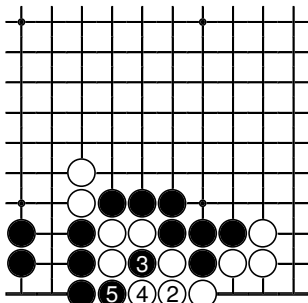
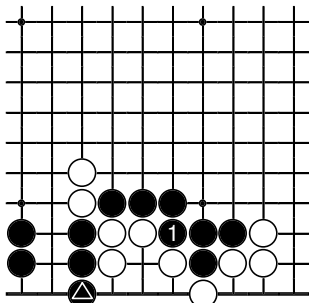
The solid connection Black 1 here is a relative failure. Black can only jump as far as 3.



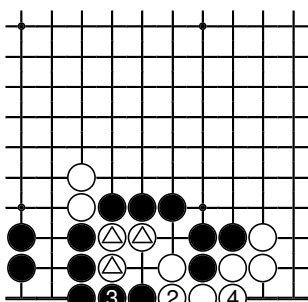
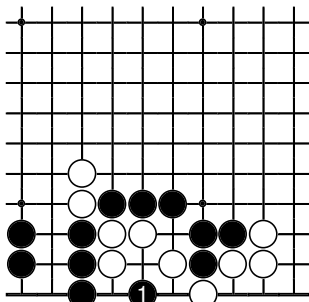
By playing B Black can capture the two cutting stones.



To play *atari* (answer A) here is a clear failure.

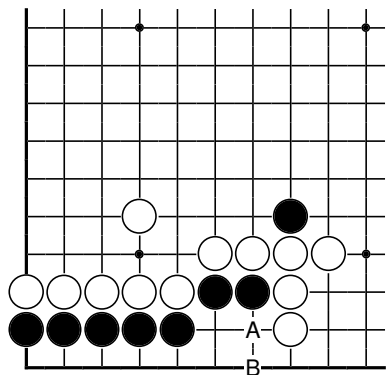
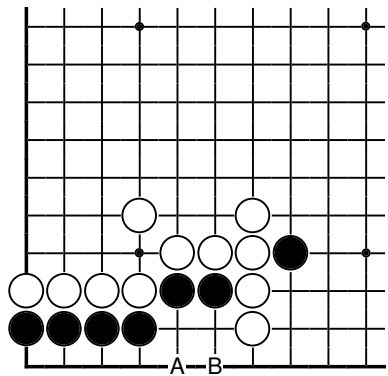


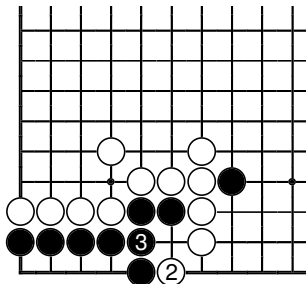
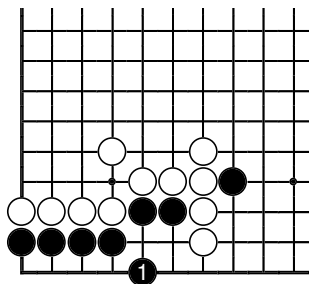
Choice B is correct, making use of the marked stone to capture the cutting stone. (This well-hidden sort of play, creating a cutting point from a diagonal play, is called *atekomi*, “aim inside”, in Japanese.)



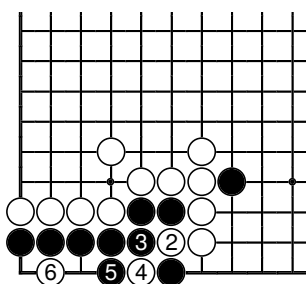
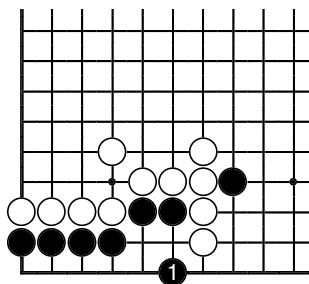
The placement Black A, at the key point for eye shape, has the disadvantage that it doesn't actually work to capture anything.

Both problems Black to play at A or B

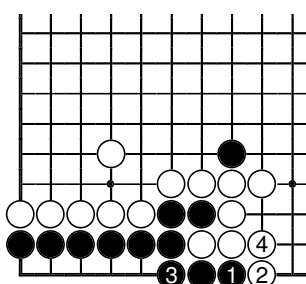
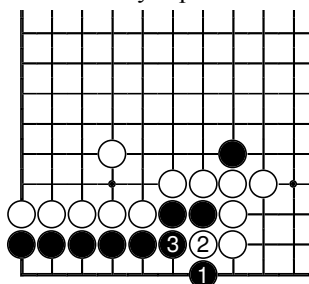




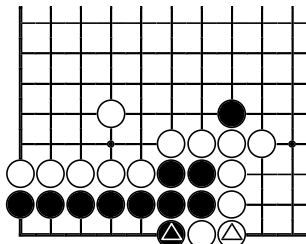
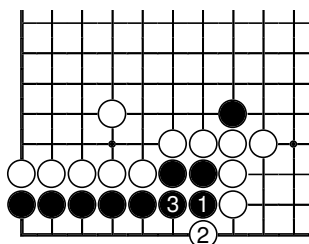
The hanging connection (answer A) is the way for Black to live here, and save two stones. When White plays 2 Black plays 3 to avoid the snapback.



The one-point jump to the edge fails. Black doesn't have room for two eyes after White plays 2. Of the four ways to defend the two stones, answer A makes the most eye space.



This problem is about correct endgame play, not life-and-death. In this case answer B, the one-point jump, takes most points. **(Right)** Black expects to play here in *sente*.

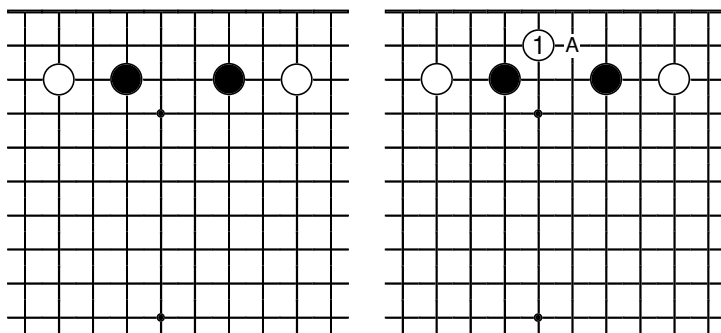


Blocking at 1 allows White 2. White will have two extra points of territory, compared to Black's correct answer.

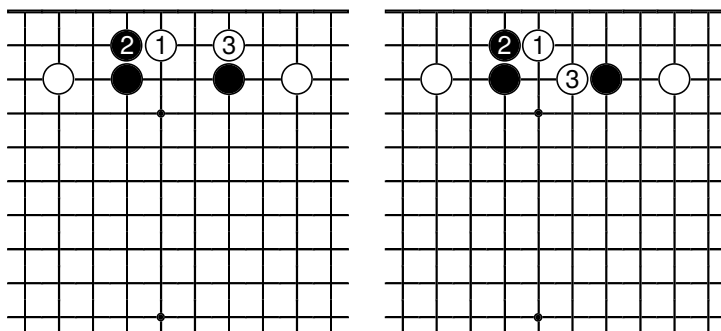
Chapter Eleven

Cramp

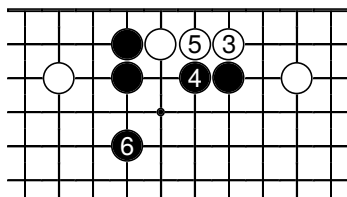
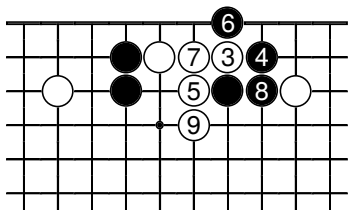
11.1 Two-point extension: the placement



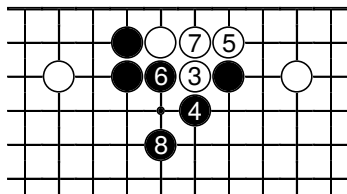
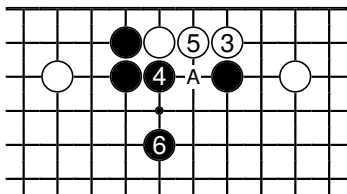
This chapter gives the other side of the story on the two-point extension. When it is cramped by two White stones, as shown in the left-hand diagram, it can be attacked in many ways. The placement (**right**) at 1 or A is something of a revelation, when you first discover it.



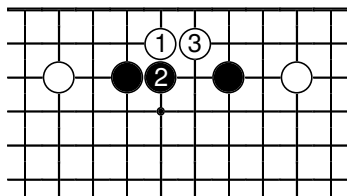
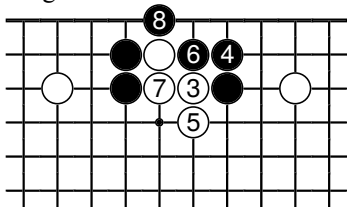
Next if Black blocks at 2, White should play 3 in the left-hand diagram; the other choice (**right**) can be criticised.



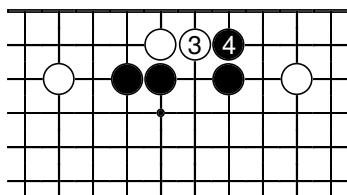
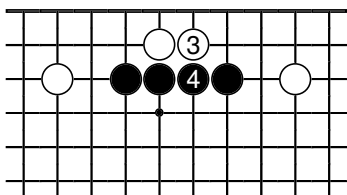
Black can't hope to resist as in the left-hand diagram. What about the right-hand way?



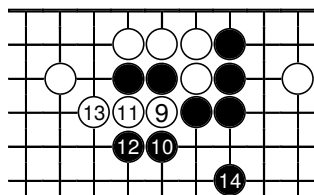
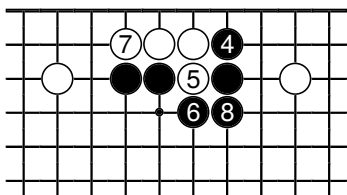
(Left) Black makes better shape with this choice of 4. White should just play 5, rather than A. **(Right)** This way of playing 3 is a little vulgar; the exchange of 3 and 4 benefits Black.



There is also a chance Black will resist **(left)**, leading to a wild position, both having bad shape. **(Right)** If Black plays 2 White 3 is inevitable.



(Left) White can connect out either side, but is thin. **(Right)** Black blocks with the intention of sacrificing if necessary.



White 5 here is reasonable. But cutting at 9 just helps Black make good shape up to 14. White should attack on a larger scale than this.

Part Five

Theory

Chapter Thirteen

Theory Applying to Effective Play

13.1 Doing the necessary, or losing the plot?

The central character in Pushkin's *Queen of Spades* is led to his downfall by overriding his cherished principle, of 'not risking the necessary to gain the superfluous'. In go, it is often hard to understand how to distinguish the two. One aspect of improving your strength is to shed all unnecessary plays. In a sense this is more important than making good shape. The gain in making the correct shape may only be a couple of points, compared with the second-best play. Some misdirected moves are almost completely wasted.

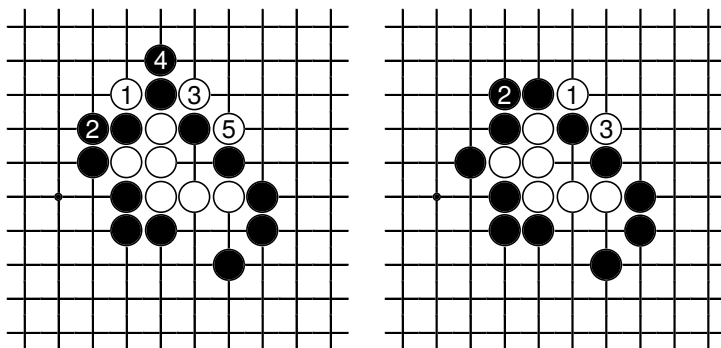
It is common to characterise inferior moves as 80% or 50% of full value, and so on. An amateur 1 *dan* probably plays 90% moves, on average over the game; top players operate at around 98% or 99% efficiency. These figures do assume that perfect play in go resembles what can be seen in professional go. This is simply a hypothesis, extrapolating from the current state of knowledge, and it is hard to see how to test it.

What are the most common causes of wasted plays? High on the list are:

- playing to save a group that is already alive;
- threatening a group with a play that isn't in fact *sente*;
- capturing stones that have no strategic meaning;
- defending territory that is badly located (e.g. open-skirted);
- carrying on in a set sequence through momentum alone.

General classes of mistakes are *safety plays* (nothing safe about playing badly), *miscalculations about the burden of proof* (if a forcing play isn't clearly forcing and clearly required right now, it is quite likely to be bad), and *misconceptions* about which are the key areas or stones.

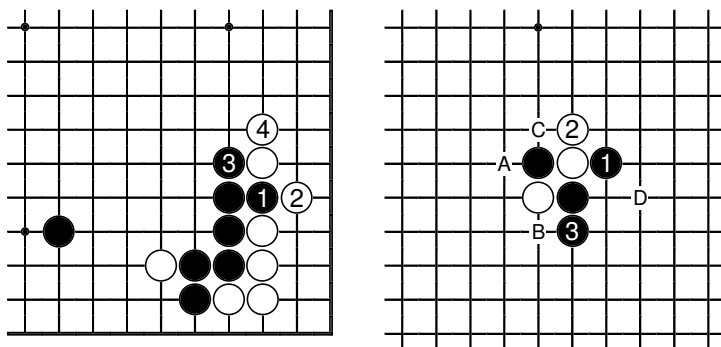
13.2 123 and use of threats



Both of these sequences allow White to escape. But in the left-hand diagram Black gains more outside influence. Since the exchange of White 1 for Black 2 in that diagram isn't necessary, it should be omitted.

123 Principle

Don't play 1-2-3, just play at 3.

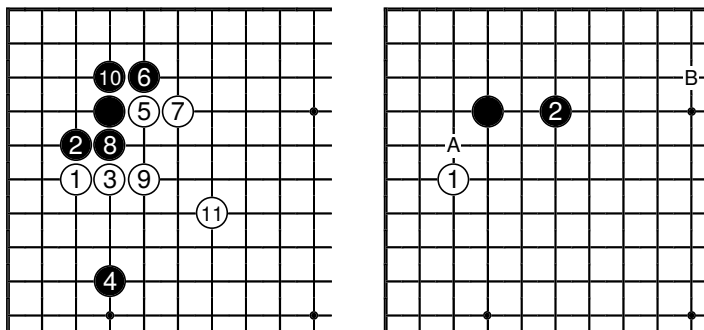


(Left) There is no good reason to play 1 here. It loses Black a liberty, and a *ko* threat; and also some potential for later play with Black at 2 or one point above. **(Right)** A cross-cut: see p.103 for Black 3 connecting solidly after Black 1 *atari*. A very common case of the principle is: 'don't play *atari* and then connect'. That can look like planless play. Black 3 in this diagram is part of a plan, to sacrifice one stone, with White A, Black B, White C, Black D. That builds strong shape for Black in one direction. Quite generally, if your play 1 has an obvious answer, you should have play 3 already lined up.

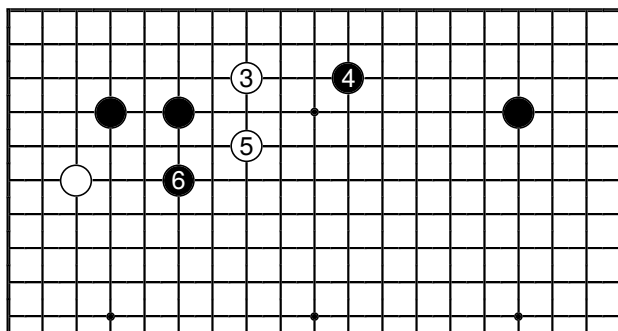
One of the key proverbs is ***don't go back to patch up***. Effective play normally aims to generate sustainable forward momentum.

13.4 Double-purpose plays

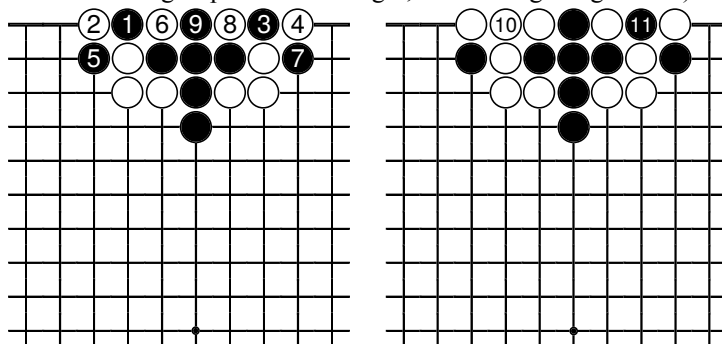
Killing two birds with one stone is a proverb in many languages. If you want your stones to work harder for you, place them where their purpose in life (or death) isn't limited to just one future direction of play.



In the left-hand diagram Black attacks White single-mindedly. On the right Black 2 sets up subsequent plays at A to attack, or B to build a framework. This is more reasonable. The points A and B are like *miai* (13.3).



If White challenges Black on the upper side with 3, Black 4 has a double purpose (attack White, build up the top right), and Black 6 has three aims (attack the White groups to left and right, and avoid getting shut in).



A well-masked trap in the endgame (White 4 should be at 7). Black 9 sets up two *kos*, both dangerous to White (who has to find the first *ko* threat).